9 Operational Amplifier

INTRODUCTION

- Linear integrated circuits are being used in a number of electronic applications such as in field like audio and radio communication, medical electronic, instrumentation control, etc. An important linear IC is operational amplifier.
- An Op-Amp is a differential amplifier having very high gain as well as very low output impedance and very high input impedance. Op-Amp are used in various applications, such as oscillators, active filter design, clipper and clamper and many types of instrumentation amplifier circuits. Op-Amp has very high voltage gain because it uses a number of differential amplifiers.

Rack Your Brain

Op-Amp is called the operational amplifiers?

Grey Matter Alert!

In Op-Amp, input of one terminal is inserted into the input of another terminal, therefore, it is also known as direct coupled amplifier.

Block diagram representation of a typical Op-Amp:

The below given diagram is the block diagram of a typical Op-Amp. In the below diagram the name of each block and biasing terminal are indicated.

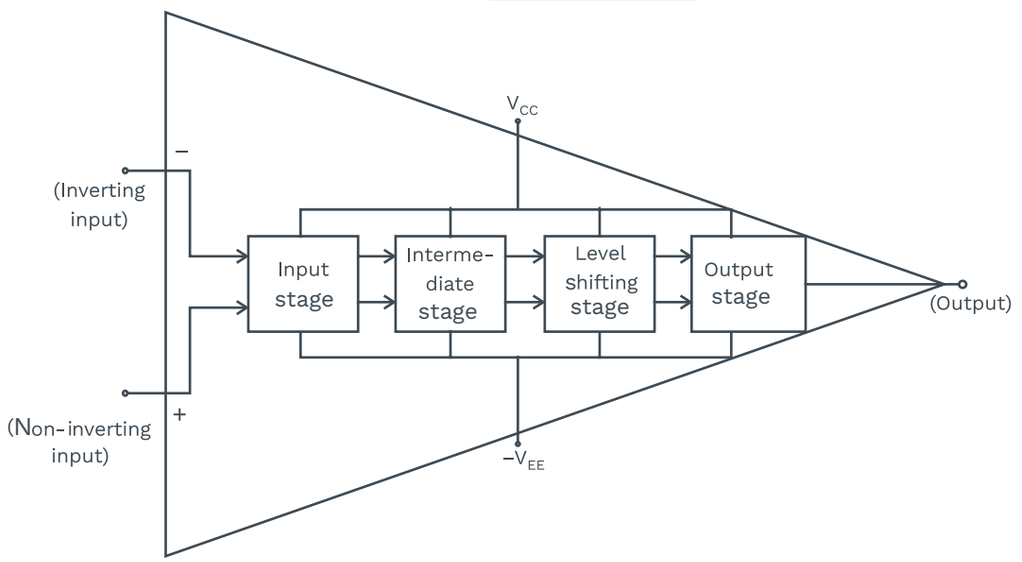


Fig. 9.1 Equivalent Circuit of Op-amp

Now, we will study in detail about every block and what is the function of each block. A simplified Op-Amp block diagram is given below with some important information about each block.

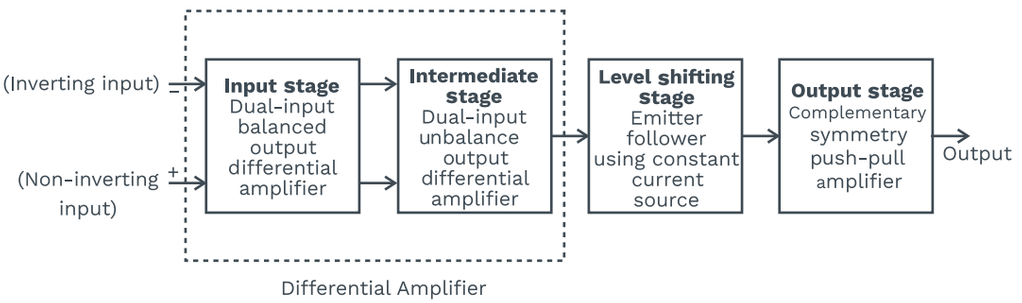


Fig. 9.2 Block Diagram of Op-amp

Explanation:

Input stage: Input stage is used to create the difference between two input signal. It also provides high input impedance to the amplifier which is one of the necessary parameter of the amplifier. You can also see from the above figure that dual input balanced output differential amplifier is used to increase the voltage for the next stage operation. As we know that main function of Op-Amp is that, first Op-Amp is used to create the difference between input signal and thereafter differentiated signal is amplified.

Intermediate stage: Output of input stage acts as an input to the intermediate stage. Direct coupling is used in this stage so that DC voltage is greater than ground potential.

Level shifting stage: Output of intermediate stage act as an input to the level shifting stage. In level shifting stage, emitter follower with a constant current source is applied. Due to stage shifting of voltage level property, it is also known as level shifting stage.

Output stage: Output of level shifting stage acts as an input to the output stage. In output stage, Push-pull amplifier is used to increase the output voltage. Push-pull amplifier also increases current delivering capability of the Op-Amp.

Grey Matter Alert!

Op-Amp amplifies the difference of input signal, that's why it is also known as high gain differential amplifier.

- Operational amplifier is a multifunctioning device which can be used for amplification purpose for AC signal as well as DC signal. Op-Amp was initially designed for mathematical purpose such as differentiation, integration, addition, subtraction or multiplication. Therefore, the name operational amplifier seems from it's original use for these mathematical operations.

Symbol of Op-Amp:

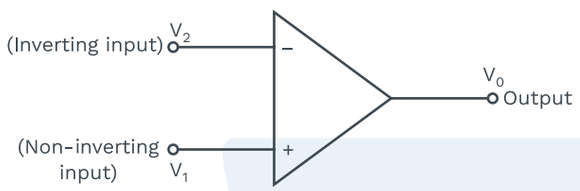


Fig. 9.3 Symbol of Op-amp

- Fig. 9.3 represent schematic symbol of an operational amplifier. The above symbol is mostly used for an operational amplifier with one output and two input.

Equivalent circuit of Op-Amp:

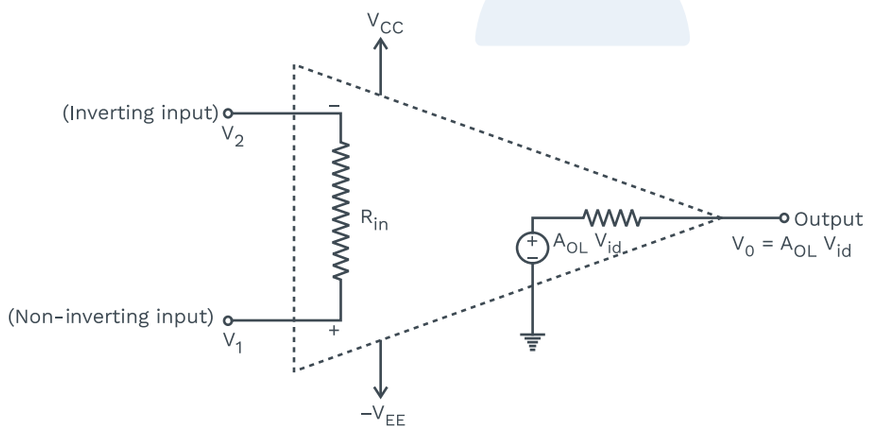


Fig. 9.4 Equivalent Circuit of Op-Amp

- Fig. 9.4 shows equivalent circuit of Op-Amp. For analyzing basic operation of Op-Amp, equivalent circuit is very useful. It also helps in observing the effects of feedback arrangements.
- Output voltage is given by:

$$V_0 = A_{OL} \times V_{id} = A_{OL} \left(V_1 - V_2\right) \qquad (9.1)$$

Where,

A_{OL} = Open-loop voltage gain

V_{id} = Input voltage difference

V_1 = Non-inverting terminal voltage

V_2 = Inverting terminal voltage

Ideal voltage transfer curve:

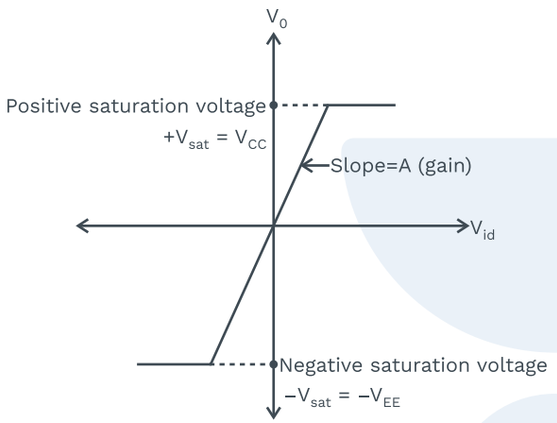


Fig. 9.5 Ideal Voltage Transfer Curve

- From equation (9.1) we will assume that output offset voltage is zero for Op-Amp. As we can see from Fig. 9.5 that output voltage cannot exceed the value of saturation voltage. For positive saturation it is $+V_{sat}$ and for negative saturation it is $-V_{sat}$. Output voltage swing rating of Op-Amp decide the saturation voltage for a specified supply voltage.
- Therefore, we can say that output voltage is directly proportional to differential input voltage upto saturation voltage and after that output voltage remains constant and equal to saturation voltage (as we can see in Fig. 9.5).

Ideal Op–Amp:

- Ideal Op-Amp is an extension of the concept of an ideal amplifier. It has zero output impedance, infinite input impedance and infinite gain. In an ideal Op-Amp, current through input terminal is zero.

Rack Your Brain

Op-Amp is also known as direct coupled high gain differential amplifier, Why?

Grey Matter Alert!

Op-Amp is known as operational amplifier because it is used for performing different functions such as differentiation, addition, integration and subtraction.

Ideal Op-Amp symbol and transfer characteristics curve:

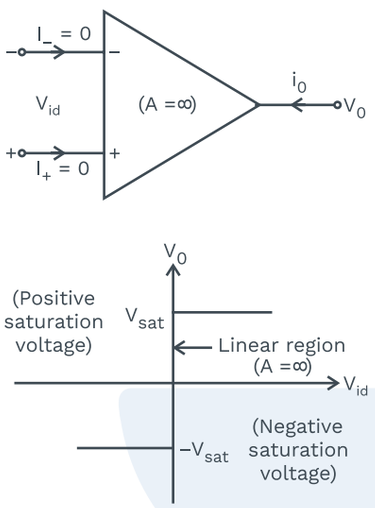


Fig. 9.6 Transfer Characteristics Curve

- While analyzing various application circuits of operational amplifiers, the integrated operational amplifier is often regarded as an ideal operational amplifier.
- The so-called ideal Op-Amp is used to idealize various technical indicators of Op-Amps, and it must have the following characteristics.

Characteristics of ideal Op-Amp:

- An ideal Op-Amp has infinite voltage gain.
- An ideal Op-Amp has infinite input resistance (R_i), because any signal can drive it and also there will be no preceding stage.
- An ideal Op-Amp has zero output resistance (R_0), because output of Op-Amp can drive an infinite number of devices.
- An ideal Op-Amp has infinite bandwidth because if signal frequency varies from zero to infinite, then also Op-Amp can amplify that signal and also without any attenuation.
- An ideal Op-Amp has Infinite common mode rejection ratio (CMRR) because output of common-mode noise voltage is zero.
- An ideal Op-Amp has infinite slew rate (SR) because if there is any change in input voltage then it must reflect to output voltage simultaneously.
- An ideal Op-Amp has zero output voltage if input voltage is zero.

Example 1.

An ideal Op-Amp can be characterized as:
a) Voltage controlled current source
b) Current controlled voltage source
c) Current controlled current source
d) Voltage controlled voltage source

Sol: d)

Ideal Op-Amp is a voltage controlled voltage source.

Example 2.

The operational amplifier use a differential input stage with a constant current source mainly to obtain:
a) Very low common mode gain.
b) Very high differential gain.
c) Very low input noise.
d) Very high input resistance.

Sol: b)

The Op-Amp use a differential input stage with a constant current source to obtain very high differential gain.

Example 3.

Consider the statements about sample and hold circuits. A sample and hold circuit has one buffer circuit at the input stage and other buffer circuit at the output stage. Now which of the following statement is correct?
a) The primary requirement for both buffer is to have high slew rate.
b) The primary requirement of input buffer is to have high slew rate and output buffer have low bias current.
c) The primary requirement for both buffer is to have low bias current.
d) The primary requirement of input buffer is to have low bias current and output buffer have high slew rate.

Sol: c)

Both the buffers should have low bias current to get the DC output voltage low.

Practical Op-Amp:

- In an ideal Op-Amp, it is assumed that the Op-Amp respond equally well to both DC and AC input voltages. However, a practical Op-Amp does not behave this way. A practical Op-Amp has some DC voltage at the output even if both the inputs are grounded.

Rack Your Brain

What are the assumptions of ideal amplifier?

- Practical Op-Amp have very high input resistance (but not infinite), gain bandwidth product is a finite quantity, consume some power and slew rate is also finite. It also have some input offset voltage as well as some input bias current.
- If we see ideal Op-Amp, then it draws no current from the source and also, response for an ideal Op-Amp is independent of temperature. However, if we see practical Op-Amp, then it does not behaves like a ideal Op-Amp. In practical Op-Amp, there is some finite amount of current through source terminal and also two input terminal of Op-Amp behaves differently to voltage and current because of mismatch of transistors. A practical Op-Amp also fluctuate or vary from its operation, if temperature is changed.

Previous Years' Question

An ideal Op-Amp is used to make an inverting amplifier. The two input terminal of the Op-Amp are at the same potential because:

a) The two input terminals are directly shorted internally.
b) The input impedance of the Op-Amp is infinite.
c) The open-loop gain of the Op-Amp is infinity.
d) CMRR is infinity.

DC characteristics of practical Op-Amp:

Some of the DC characteristics of a practical Op-Amp are:

i) Input bias current.
ii) Input offset current
iii) Input offset voltage
iv) Thermal drift

i) Input bias current:

Input bias current is defined as the average current flowing through both input terminals of Op-Amp that is non-inverting terminal as well as inverting terminal.

$$I_B = \frac{\left(I_{B_1} + I_{B_2}\right)}{2} \qquad (9.2)$$

Where,

I_{B_1} = Non-inverting terminal input current.
I_{B_2} = Inverting terminal input current.

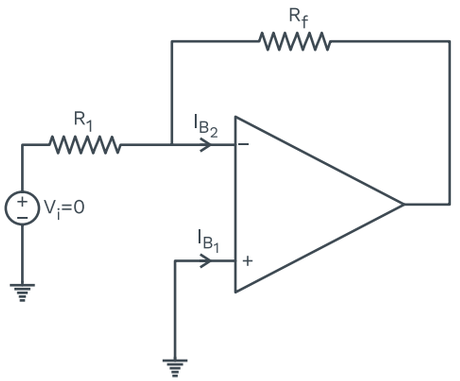


Fig. 9.7 Inverting Amplifier with Bias Current.

ii) Input offset current:

Input offset current is defined as the modulus of algebraic difference between current of inverting terminal and non-inverting terminal. It is denoted as (I_{io}).

$$I_{io} = \left| I_{B_1} - I_{B_2} \right| \qquad (9.3)$$

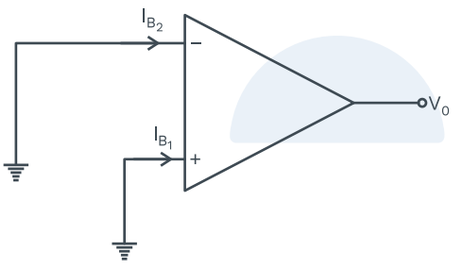


Fig. 9.8 Input Offset Current

- For IC-741 maximum input offset is 200 nA. If we want to decrease the value of input offset current (I_{io}), then we have to improve the matching between two input terminals.

iii) Input offset voltage:

Input offset voltage is defined as the voltage needed between two input terminals so that voltage at output terminal becomes zero. As we can see from Fig. 9.9. For better match between input terminals, value of input offset voltage is as small as possible.

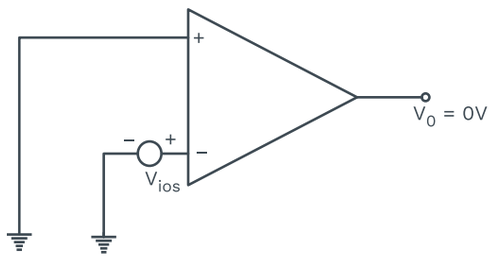


Fig. 9.9 Op-Amp Showing Input Offset Voltage

Let us now examine the effect of V_{ios} on the output of non-inverting and inverting Op-Amp as shown in Fig. [9.10a), 9.10b)].

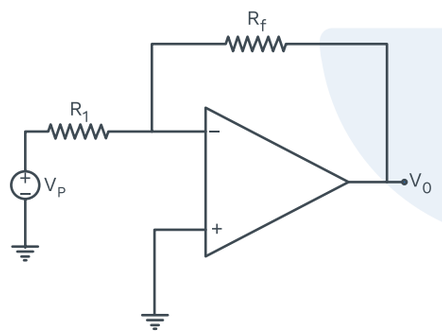


Fig. 9.10 a) Inverting Op-Amp

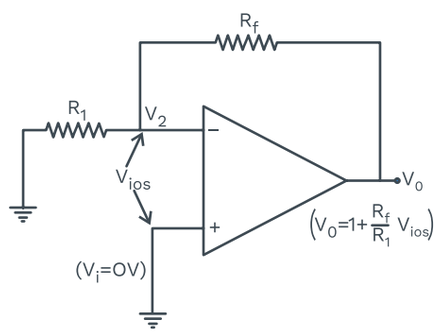
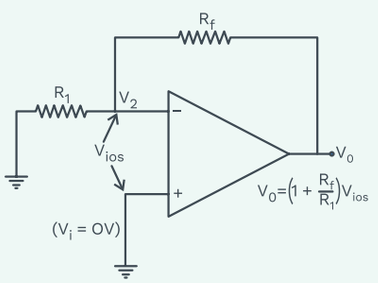


Fig. 9.10 b) Non-inverting Op-Amp

Previous Years' Question

For the circuit shown below, taking the op-amp as ideal, the output voltage V_{out} in terms of the input voltage V_1, V_2 and V_3 is:



a) $1.8\ V_1 + 7.2\ V_2 - V_3$
b) $8\ V_1 + 8\ V_2 - 9V_3$
c) $7.2\ V_1 + 1.8\ V_2 - 9V_3$
d) $8\ V_1 + 2\ V_2 - 9V_3$

If V_p is set to zero, the circuit of Fig. 9.10a) becomes the same as in Fig. 9.10b).

From Fig. 9.10b), voltage at inverting terminal is given by:

$$V_2 = \left(\frac{R_1}{R_1 + R_f}\right) \cdot V_0 \qquad (9.4)$$

or, $V_0 = \left(\frac{R_1 + R_f}{R_1}\right) \cdot V_2$

$$V_0 = \left(1 + \frac{R_f}{R_1}\right) \cdot V_2 \qquad (9.5)$$

Since $V_{ios} = |V_i - V_2|$ and $V_i = 0$ (9.6)

$$V_{ios} = |0 - V_2| = V_2$$

or $V_0 = \left(1 + \frac{R_f}{R_1}\right) V_{ios}$ (9.7)

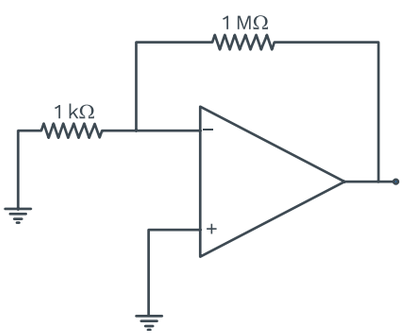
Thus the output offset voltage of an Op-Amp in closed-loop configuration (inverting or non-inverting) is given by equation (9.7).

Thermal drift:

Offset current, bias current and offset voltage generally changes with temperature. A circuit generally does not remain constant with change in temperature and this is called drift. When there is drift in current, the offset current drift is expressed in nA/°C whereas, if there is drift in voltage, then offset voltage drift is expressed as mV/°C. This indicates change in offset for change in temperature in one degree.

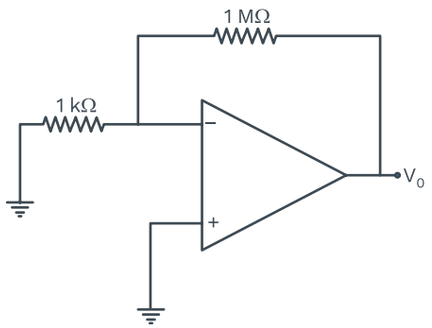
Example 4.

Given an Op-Amp is ideal but it has an input offset voltage of 1mV. Find the output voltage of Op-Amp if Op-Amp is used in the configuration shown below.



a) 1 mV **b)** 1 V **c)** ± 1 V **d)** 0 V

Sol:



Input offset voltage = V_{io} = 1mV

$= |V_+ - V_-|$

V_{io} may be negative or positive depending on the mismatch between the two inputs V_+ and V_-.

Applying KCL at inverting input, we get:

$$\frac{V_-}{R_1} = \frac{(V_0 - V_-)}{R_F}$$

$$V_0 = \frac{(R_1 + R_F)}{R_1} V_-$$

Also,

$$\therefore V_{i0} = |V_+ - V_-| = V_- \quad [\because V_+ = 0]$$

$$\therefore V_0 = \left(1 + \frac{R_F}{R_1}\right)(\pm V_{i0})$$

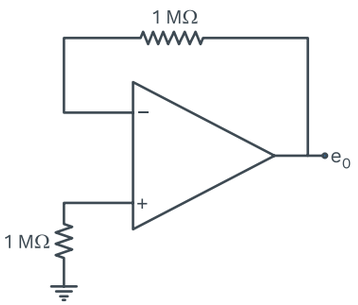
$$V_0 = \left(1 + \frac{10^6}{10^3}\right) \times (\pm 10^{-3})\ V$$

$$\therefore V_0 = \pm 1001 \times 10^{-3}\ V$$

$$\therefore V_0 = \pm 1V$$

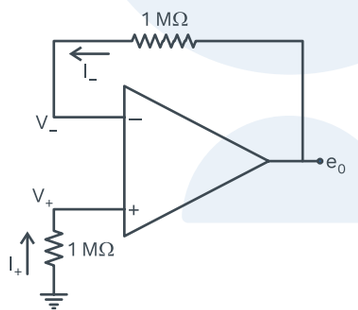
Example 5:

Output voltage of given Op-Amp is denoted by e_0 as shown in the Fig.. e_0 can be measured with the help of ideal voltmeter. Which of the following can be calculated?



a) Inverting terminal input and non-inverting terminal input only.
b) Bias current only.
c) Input offset current only.
d) Input offset current as well as bias current.

Sol:
Let us assume, gain of the Op-Amp is ∞ (i.e., ideal Op-Amp)



Applying KVL at upper loop, we get:

$$e_0 - I_- \times 1\,M = V_- \quad (1)$$

Applying KVL at lower loop, we get:

$$V_+ + V_- \times 1\,M = 0$$

$$\Rightarrow V_+ = -I_+ \times 1M \quad (2)$$

From virtual ground concept,

$$V_+ = V_-$$
$$e_0 - I_- \times 1\,M = -I_+ \times 1\,M$$
$$e_0 = I_- \times 1\,M - I_+ \times 1\,M$$
$$e_0 = (I_- - I_+) \times 1\,M$$

e_0 depends upon input offset current only.

Concept of virtual ground (VGC):
A virtual ground is a node of circuit that is maintained at a steady reference potential (nearly 0 V) without being connected directly to reference potential.

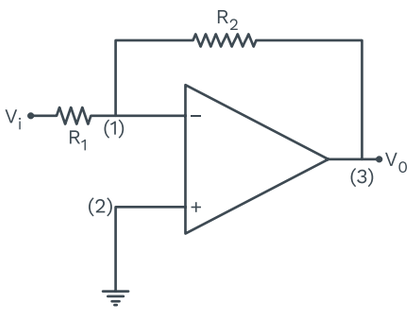


Fig. 9.11 Virtual Ground Concept

By analyzing the circuit shown in Fig. 9.11 we will see the concept of virtual ground. Equivalent circuit of Fig. 9.11 is given in Fig. 9.12. Now, we will find closed-loop voltage gain and it is defined as ratio of output voltage to input voltage.

$$A_v = \frac{V_0}{V_i} \qquad (9.8)$$

As we know that the value of open-loop gain (A_V) for an ideal Op-Amp is infinite, we will assume circuit is working properly as well as producing finite output voltage at output terminal (terminal 3). Above two condition can satisfy simultaneously if voltage difference between inverting terminal and non-inverting terminal should be negligible or ideally zero. Therefore, we can write:

$$V_2 - V_1 = \frac{V_0}{A_V} = 0$$

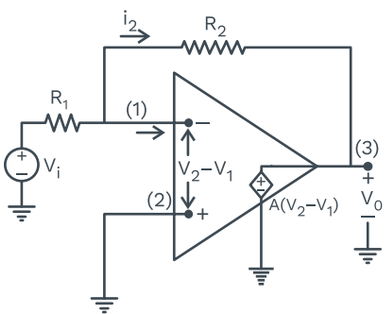


Fig. 9.12 Virtual Ground Concept

From the above equation, we can say that voltage at inverting terminal equal to non-inverting terminal because open-loop gain (A_V), approaches to infinity. Now we can say that two input terminal are tracking each other in potentials. We can also say that two input terminals are virtual short circuit. Virtual should be emphasized because there is no direct connection between the two input terminals.

- Virtual short means whatever the voltage at one terminal the same voltage will appear across another terminal because of infinite open -loop gain (A_V). Virtual ground means if one terminal either non-inverting terminal or inverting terminal is grounded then automatically another terminal will be grounded by the concept of virtual short that is if V_2 = 0 then V_1 = 0 or reverse.

Inverting amplifier:

- Inverting amplifier is a type of operational amplifier in which output produced by amplifier is 180° phase shift w.r.t. input signal. If input signal is negative then output signal will be positive or if input signal is positive, then output signal will be negative. Circuit diagram of inverting amplifier by using Op-Amp is shown in Fig. 9.13.

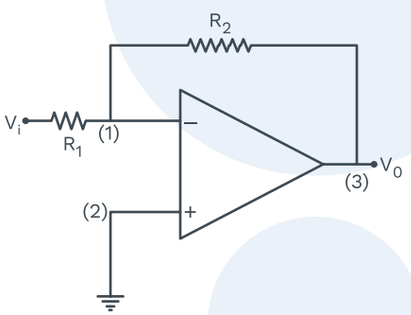


Fig. 9.13 Inverting Amplifier

We say that, if R_2 is connected between (3) and (1), then it is negative feedback and if it is connected between (3) and (2), then it is positive feedback.
Also R_2 closes the loop around Op-Amp.
So, we will analyze the circuit now:

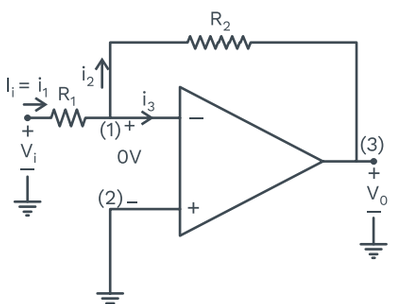


Fig. 9.14 Inverting Amplifier

Due to virtual ground concept,

$$V_2 = V_1 \quad (9.9)$$

so, $V_2 = 0$ V

So, $V_2 = V_1 = 0$ V (9.10)

Current through inverting terminal is, approximately zero, i.e. $i_3 \approx 0$

So,

$$I_i = \frac{(V_i - V_1)}{R_1} \quad (9.11)$$

(From equation (9.10))

$$I_i = \frac{(V_i - 0)}{R_1}$$

$$I_i = \frac{V_i}{R_1} \quad (9.12)$$

Now, applying KCL at node 1.

So,

$$\frac{(V_i - 0)}{R_1} = \frac{(0 - V_0)}{R_2} \quad (9.13)$$

$$V_0 = \frac{-R_2}{R_1} V_i \quad (9.14)$$

Now, closed-loop voltage gain will be:

$$A_V = \frac{V_0}{V_i} = \frac{-R_2}{R_1} \quad (9.15)$$

Here, in gain, because of the minus sign associated with close-loop gain, this configuration is called inverting configuration.

Calculation of input resistance:

- We can determine the input resistance seen by the voltage source V_i.
- Due to virtual ground we have from equation (9.12).

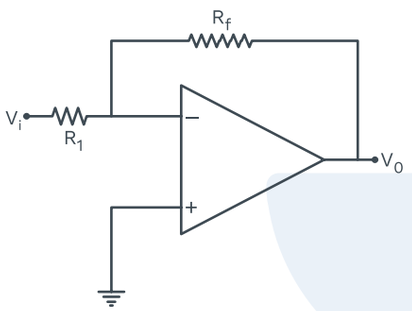
$$I_i = \frac{V_i}{R_1} \quad (9.16)$$

Input resistance is defined as:

$$R_i = \frac{V_i}{I_i} = R_1 \quad (9.17)$$

So, now from equation (9.17), we can see that the input resistance seen by the source is a function of R_1 only and is a result of the 'virtual ground' concept.

Example 6:
An inverting amplifier circuit is given below. Find the value of output voltage if R_1 = 100 kΩ, R_f = 500 kΩ and V_i = 2 V?



Sol:
From equation (9.14):
We, have

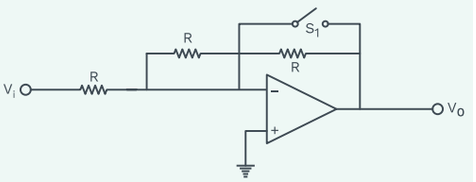
$$V_0 = -\frac{R_f}{R_1} \times V_i$$

$$V_0 = -\frac{500}{100} \times 2 \text{ V}$$

$$V_0 = -10 \text{ V}$$

Previous Years' Question

Let the magnitude of the gain in the inverting Op-Amp circuit shown be x with switch S_1 open. When the switch is closed, the magnitude of gain becomes:



a) x/2 **b)** –x
c) 2x **d)** –2x

Effect of finite open-loop gain:

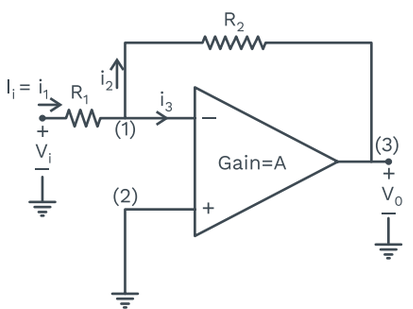


Fig. 9.15 Effect of finite open–loop gain

- We will understand in more clear way if we assume open-loop gain finite and then derive the expression of gain in an Op-Amp. We will derive gain expression by using Fig. 9.15. Now current through inverting terminal that is (i_3) is equal to zero ($i_3 = 0$).
- Now output voltage is denoted as V_0 then voltage between inverting terminal and non-inverting terminal will be equal to $\frac{V_0}{A}$. Now as we can see from Fig. 9.15 that non-inverting terminal is grounded then voltage at inverting terminal must be equal to $-\frac{V_0}{A}$.

So,
Now current I_i through R_1 can be found from:

$$I_i = \frac{V_i - \left(-V_0 / A\right)}{R_1} \qquad (9.18)$$

$$I_i = \frac{V_i + \left(\frac{V_0}{A}\right)}{R_1} \qquad (9.19)$$

The infinite input impedance of Op-Amp forces the current I_i to flow entirely through R_2.
The output voltage V_0 can thus be determined from ($I_i = i_1 = i_2$),

$$V_0 = -\frac{V_0}{A} - i_2 R_2 \qquad (9.20)$$

$$V_0 = -\frac{V_0}{A} - \frac{\left(V_i + V_0/A\right)}{R_1} \times R_2 \qquad (9.21)$$

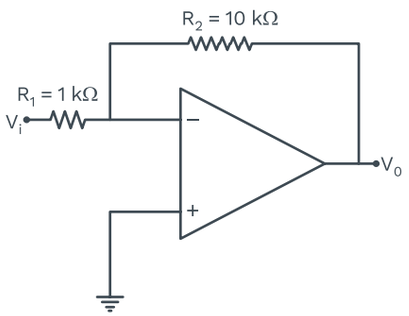
Collecting terms, the closed-loop gain A_V is found to be:

$$A_V = \frac{V_0}{V_i} = \frac{-R_2 / R_1}{1 + \frac{\left(1 + R_2 / R_1\right)}{A}} \qquad (9.22)$$

In equation (9.22), it is shown that if $A \rightarrow \infty$, then the ideal closed-loop voltage gain reduces and that is given by equation (9.15).

Example 7:

An inverting amplifier is given below. What will be the value of closed-loop gain that is $\frac{V_0}{V_i}$, if open-loop gain is 100?



Sol:

From equation (9.23), we have seen that:

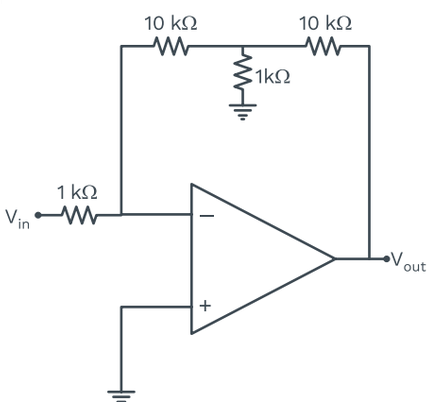
$$A_V = \frac{V_0}{V_i} = \frac{-\left(R_2 / R_1\right)}{1 + \frac{\left(1 + R_2 / R_1\right)}{A}}$$

$$A_V = \frac{-10 / 1}{1 + \frac{(1 + 10 / 1)}{100}}$$

$$A_V = -9$$

Example 8:

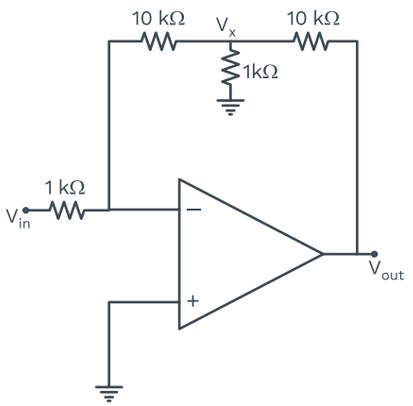
An inverting operational amplifier is given below. What will be the value of closed-loop gain $\frac{V_{out}}{V_{in}}$?



Sol:

Let voltage at input terminals of Op-Amp are V_- and V_+ respectively.

So, $V_+ = V_- = 0$



Applying KCL at inverting terminal:

$$\frac{(0 - V_{in})}{1} + \frac{(0 - V_x)}{10} = 0 \qquad (1)$$

At node x:

$$\frac{(V_x - 0)}{10} + \frac{(V_x - V_0)}{10} + \frac{(V_x - 0)}{1} = 0$$

$$V_x + V_x - V_0 + 10V_x = 0$$

$$12V_x = V_0$$

$$V_x = \frac{V_0}{12} \qquad (2)$$

Now, from equation (1):

$$\frac{V_{in}}{1} + \frac{V_x}{10} = 0$$

$$V_{in} = \frac{-V_{out}}{120}$$

$$\frac{V_{out}}{V_{in}} = -120$$

Non-inverting amplifier:

- Non-inverting amplifier is a type of operational amplifier in which output produced by amplifier is in phase with input. Therefore, we can say phase difference between input and output signal is either 0° or 360°. If input signal is negative then output signal will be negative or if input signal is positive then output signal will be positive. In non-inverting amplifier, input is fed at non-inverting terminal of Op-Amp. Negative

feedback is also used by non-inverting amplifier but instead of giving entire input signal to feedback, in this configuration only some portion of input signal is given as a feedback connection. Circuit diagram of non-inverting amplifier by using Op-Amp is shown in Fig. 9.16.

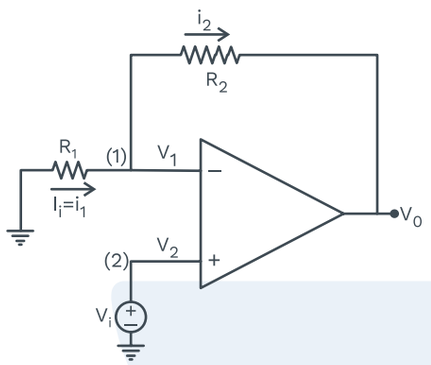


Fig. 9.16 Non-inverting Amplifier

In Fig. 9.16 we can see that input voltage is applied at non-inverting terminal whereas at inverting terminal resistor R_1 is connected and the other side of resistor is grounded. Analysis of non-inverting amplifier is almost similar to inverting amplifier. We will assume that there is no current through input terminal of Op-Amp.

Since, $V_1 = V_2$, then, $V_1 = V_i$ and current I_i is given by:

$$I_i = \frac{(0 - V_i)}{R_1} = \frac{-V_i}{R_1} \qquad (9.24)$$

Current i_2 is given by:

$$i_2 = \frac{V_1 - V_0}{R_2} = \frac{V_i - V_0}{R_2} \qquad (9.25)$$

As $i_1 = i_2$, so:

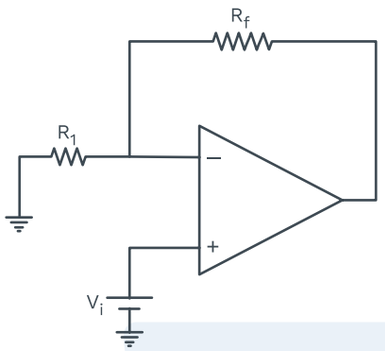
$$-\frac{V_i}{R_1} = \frac{V_i - V_0}{R_2} \qquad (9.26)$$

While we solve for closed-loop voltage gain, we find that

$$A_V = \frac{V_0}{V_i} = 1 + \frac{R_2}{R_1} \qquad (9.27)$$

- From the above equation, we can see that, the output is in phase with input. Gain is always greater than 1 (gain>1)

Example 9:
Given below is a non-inverting amplifier. What will be the value of output voltage if R_f = 500 kΩ, R_1 = 100 kΩ and V_i = 2 V?



Sol:
From equation (9.27), we can see that,

$$A_V = \frac{V_0}{V_i} = \left(1 + \frac{R_f}{R_1}\right)$$

$$V_0 = \left(1 + \frac{500}{100}\right) \times 2 \text{ V}$$

$$V_0 = 12 \text{ V}$$

Voltage follower circuit:

- A voltage follower circuit is a special case of non-inverting amplifier in which feedback resistance (R_f) is zero and value of resistance (R_1) is infinite. In this, output voltage follows input voltage which means output voltage is same as input voltage, that's why it is known as voltage follower circuit.

Rack Your Brain

What is voltage follower circuit and what are it's advantages?

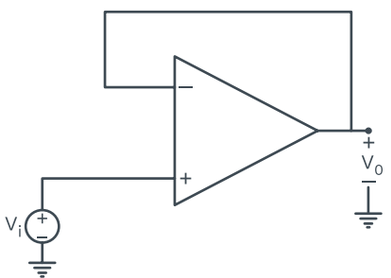


Fig. 9.17 Voltage Follower Circuit

$$V_0 = V_i \qquad (9.28)$$

The voltage-follower Op-Amp circuit is shown in Fig. 9.17. The property of high input impedance is a desirable feature of non-inverting configuration. which enables this circuit to behave as a buffer amplifier to connect a source with high impedance to low-impedance load.

- Finally gain of buffer amplifier is unity, i.e.

$$A_V = \frac{V_0}{V_i} = 1 \qquad (9.29)$$

Example 10:

Consider the statement about voltage follower circuit design using Op-Amp. Which of them is correct?

a) Input impedance of voltage follower circuit is very small whereas output impedance of voltage follower circuit is very high.

b) Input impedance of voltage follower circuit is very high whereas output impedance of voltage follower circuit is very low.

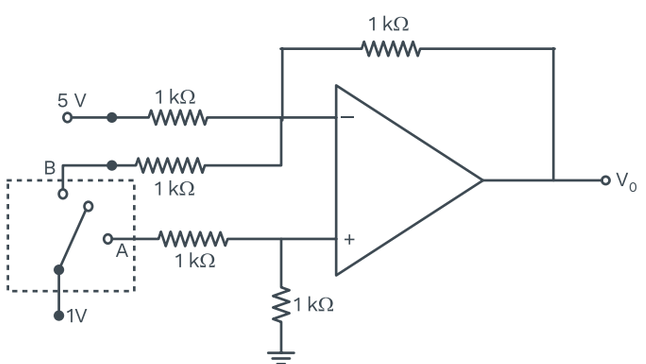
c) Input impedance as well as output impedance of voltage follower circuit is very high.

d) Input impedance as well as output impedance of voltage follower circuit is very small.

Sol: Input impedance is very large and the output impedance is very small.

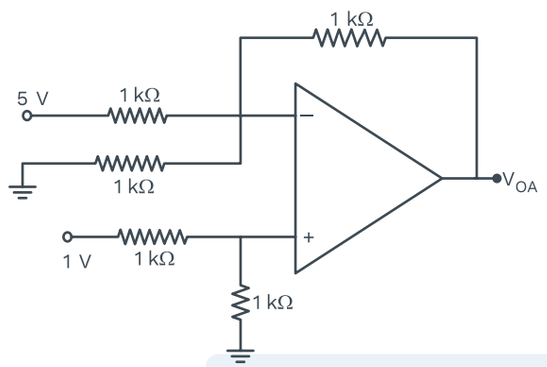
Example 11:

In the circuit shown $V_0 = V_{OA}$ for switch in position A and $V_0 = V_{OB}$ for Switch in position B. Assume that the Op-Amp is ideal. The value of $\frac{V_{OB}}{V_{OA}}$ is__.



Sol:

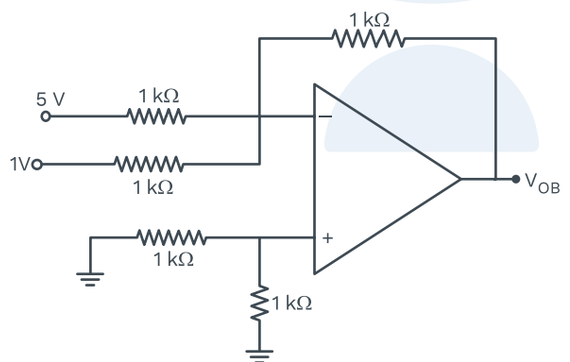
When switch is in position A:



$$V_{OA} = -5\left(\frac{1}{1}\right) + 1\times\frac{1}{2}\times\left(1+\frac{1}{1}+\frac{1}{1}\right)$$

$$V_{OA} = -3.5 \text{ V}$$

When switch is in position B:

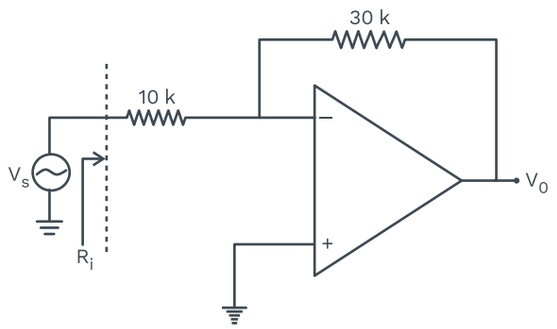


$$V_{OB} = -5\times\left(\frac{1}{1}\right) - 1\times\left(\frac{1}{1}\right)$$

$$V_{OB} = -6\text{V}$$

$$\therefore \frac{V_{OB}}{V_{OA}} = \frac{-6}{-3.5} = 1.7$$

Example 12:
The input resistance R_i of the amplifier shown in the Fig. is:



Sol:

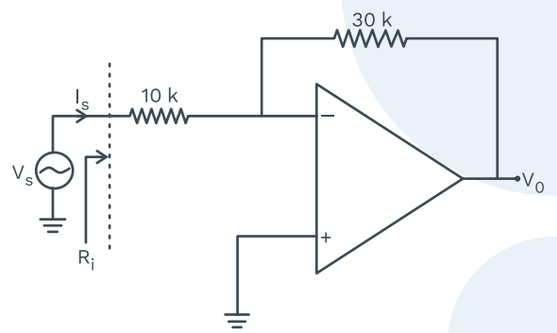


Fig. 9.31

Let inverting terminal be A.
By virtual short

$$V_A = 0$$

Now,

$$V_S - V_A = 10k(I_S)$$

or,

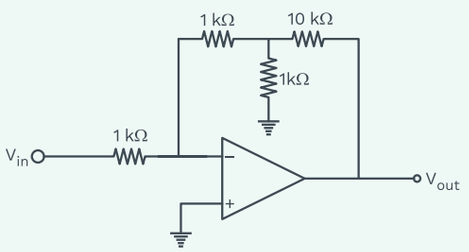
$$\frac{V_S}{I_S} = 10k$$

So, $R_i = 10\,k\Omega$

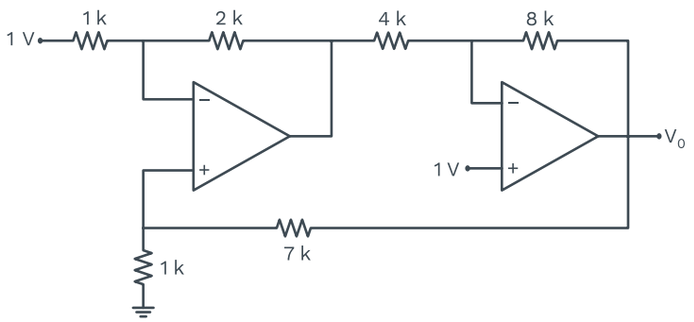
Example 13:
Find the output voltage of the following circuit assuming ideal Op-Amp behaviour.

Previous Years' Question

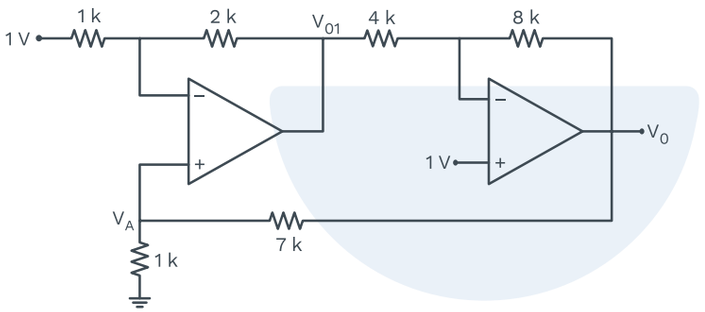
Assuming the operational amplifier to be ideal, the gain $\frac{V_{out}}{V_{in}}$ for the circuit shown in Fig. is:



a) –1 **b)** –20
c) –100 **d)** –120



Sol:



Applying KCL at node A:

$$\frac{(V_A - V_0)}{7} + \frac{V_A}{1} = 0$$

Or, $V_A = \left(\frac{V_0}{8}\right)$

Now, output of first Op-Amp.

$$V_{01} = -\frac{R_f}{R_1}V_i + \left(1 + \frac{R_f}{R_1}\right)V_A$$

or,

$$V_{01} = -\frac{2}{1} \times 1 + \left(1 + \frac{2}{1}\right) \times \frac{V_0}{8}$$

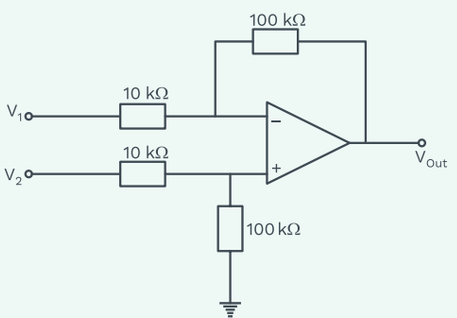
$$V_{01} = -2 + \frac{3V_0}{8}$$

Output of second Op-Amp:

$$V_0 = \frac{-R_f}{R_2} \cdot V_{01} + \left(1 + \frac{R_f}{R_2}\right) \times 1$$

Previous Years' Question

In the circuit below, the operational amplifier is ideal. If V_1 = 10 mV and V_2 = 50 mV, the output voltage (V_{out}) is:



a) 400 mV **b)** 500 mV
c) 600 mV **d)** 100 mV

$$V_0 = -\frac{8}{4}V_{01} + \left(1+\frac{8}{4}\right)\times 1$$

$$V_0 = -\frac{8}{4}\left(\frac{3V_0}{8} - 2\right) + 3$$

$$V_0 = -\frac{3}{4}V_0 + 4 + 3$$

$$\frac{7V_0}{4} = 7$$

$$V_0 = 4\ V$$

Slew rate:

Maximum rate of change in output voltage per unit time is known as slew rate (SR). It is expressed in volts per microsecond.

In equation form:

$$SR = \left.\frac{dV_0}{dt}\right|_{maximum} \quad V/\mu s \qquad (9.29)$$

- Slew rate indicates if the input frequency of Op-Amp is changed then how rapidly the output of Op-Amp changes. Generally, slew rate of Op-Amp is fixed. Distortion will occur in output signal if slope requirement of output signal is greater than slew rate.

Slew rate equation derivation:

- Generally, at the time of slew rate we will assume voltage gain as unity. Therefore, for calculation purpose, we will take voltage follower circuit as shown in Fig. 9.18. Now we will assume that amplitude and frequency of input signal is large.

Rack Your Brain

What is slew rate?

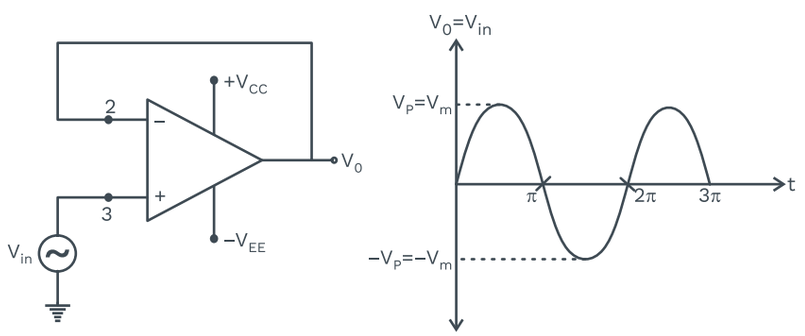


Fig. 9.18 Voltage Follower Circuit and Output Waveform

The equation for the sine wave is-

$$V_{in} = V_m \cdot \sin \omega t \qquad (9.30)$$

or $V_0 = V_m \sin \omega t \qquad (9.31)$

The rate of change of the output voltage is given by:

$$\frac{dV_0}{dt} = V_m \omega \cos \omega t \qquad (9.32)$$

and the maximum rate of change of output occurs when cos ωt = 1,
That is.

$$\left.\frac{dV_0}{dt}\right|_{max} = V_m \omega \qquad (9.33)$$

So, Slew rate (SR) = $V_m \omega$

$$SR = 2\pi f V_m \ V/s$$

$$SR = \frac{2\pi f V_m}{10^6} V/\mu \qquad (9.34)$$

Where

SR = Slew rate (V/μ)
f = Input frequency (Hz)
V_m = Peak value of the output sine wave (volts)

- If we want our output signal to be distortion-less than slew rate determines the value of maximum frequency (f_{max}) for operation of Op-Amp as well as, it also determines the desired output swing. Equation (9.34) gives the idea about distortion-less signal at output that is till the value of right side of equation (9.34) is less than left side of equation (9.34) output signal will be undistorted or distortion-less. If either of them, that is amplitude or frequency is changed, right side becomes greater than left side then output will be distorted.

Example 14:
A zero-crossing detector is made using Op-Amp. Now slew rate of Op-Amp is 12 V/μs and maximum output for the Op-Amp is ±12 V_{P-P}. Assume there is no reduction in peak to peak output then find the value of maximum frequency of input signal?

Sol:

Slew rate = $2\pi f_o V_0$

$$f_0 = \frac{(\text{slew rate})}{2\pi V_0}$$

$$f_0 = \frac{12}{10^{-6}} \times \frac{1}{2\pi \times 12}$$

$$f_0 = 159 \text{ KHz}$$

Example 15:
An amplifier is made by using Op-Amp. This amplifier is capable of faithful amplification of sinusoidal signal from DC to 20 kHz without any distortion. Given gain of amplifier is 40dB and slew rate of amplifier is 1V/μs. What will be the maximum value of input signal level?

Sol:

Slew rate $= A \cdot 2\pi f V_m$

$$V = A V_m \sin \omega t$$

$$\frac{dV}{dt} = A V_m \omega \cos \omega t$$

$$\left.\frac{dV}{dt}\right|_{max} = SR = A \cdot V_m \cdot 2\pi f_m$$

$$20 \log A = 40$$

Now, $A = 100$

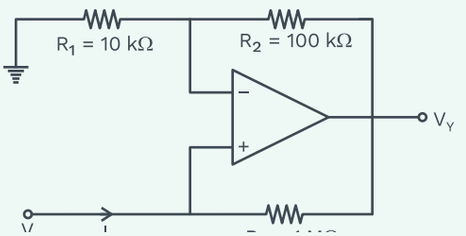
$$V_m = \frac{SR}{A \cdot 2\pi f_m}$$

$$V_m = \left(\frac{1}{10^{-6} \times 100 \times 2\pi \times 20 \times 10^3}\right)$$

$$V_m = 79.5 \text{ mV}$$

Previous Years' Question

The input resistance $\left[R_{in} = \left(\frac{V_x}{I_x}\right)\right]$ of the circuit in Fig. is:



a) + 100 kΩ **b)** – 100 kΩ
c) +1 kΩ **d)** –1 kΩ

Summing amplifier:
Summing amplifier is a special type of amplifier in which output is the sum of all the input voltage present in an amplifier.

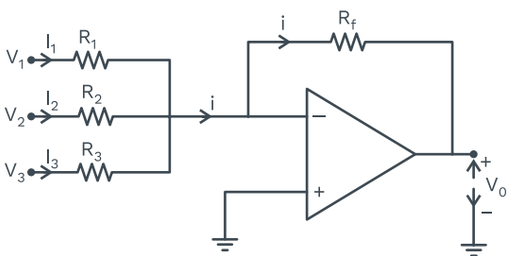


Fig. 9.19 A Weighted Summer

Fig. 9.19 shows the inverting weighted-summer circuit. From our previous concept the ideal Op-Amp will have a virtual ground appearing at its negative input terminal. Ohm's law tells us that current I_1, I_2,.., I_n are given by:

$$I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \cdots, I_n = \frac{V_n}{R_n} \tag{9.35}$$

All these currents sum together to produce the current i); that is:

$$i = I_1 + I_2 + I_3 + \cdots + I_n \tag{9.36}$$

Current i is found to flow through R_f (Since no current flows into the input terminal of an ideal Op-Amp).

The output voltage V_0 may now be determined by Kirchhoff's voltage law, i.e.

$$V_0 = 0 - iR_f \tag{9.37}$$

$$V_0 = -iR_f$$

$$\text{So, } V_0 = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \cdots + \frac{R_f}{R_n}V_n\right) \tag{9.38}$$

That is the output voltage is weighted sum of input signals V_1, V_2, ..., V_n. This circuit is therefore called weighted summer.

Special case:

$$\text{If } \quad R_f = R_1 = R_2 = \ldots = R_n \tag{9.39}$$

then,

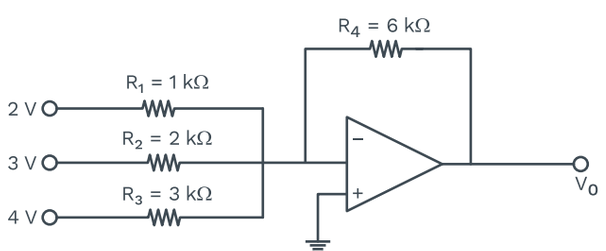
$$V_0 = -\left(\frac{R_1}{R_1}V_1 + \frac{R_2}{R_2}V_2 + \cdots + \frac{R_n}{R_n}\cdot V_n\right) \tag{9.40}$$

$$V_0 = -\left(V_1 + V_2 + \cdots + V_n\right) \tag{9.41}$$

So, it act as summing of all input voltage.

Example 16:

In the given circuit, the magnitude of output voltage is:



Sol:

Output voltage is given by,

$$v_0 = -\left(\frac{R_4}{R_1}\cdot v_1 + \frac{R_4}{R_2}\cdot v_2 + \frac{R_4}{R_3}\cdot v_3\right)$$

So, by putting value,

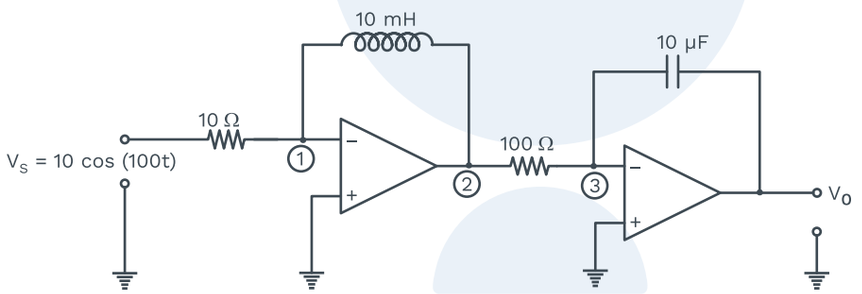
$$v_0 = -\left(\frac{6}{1}\times 2 + \frac{6}{2}\times 3 + \frac{6}{3}\times 4\right)$$

v_0 = -(12 + 9 + 8)

v_0 = – 29 v

Example 17:

In the the below given figure assume that Op-Amps are ideal. The output V_0 of the circuit is:



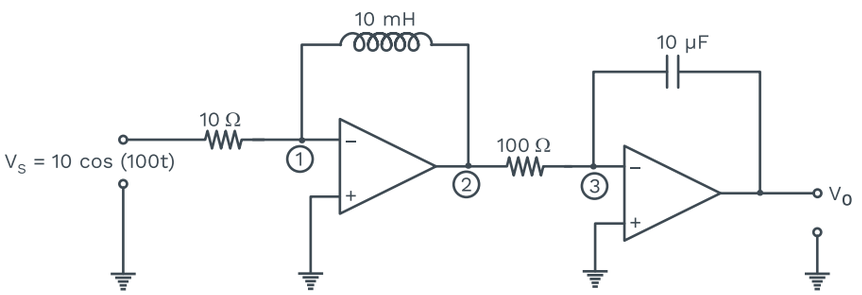
a) 10 cos (100t) V

b) $10\int_0^t \cos(100\tau)d\tau$ V

c) $10^{-4}\int_0^t \cos(100\tau)d\tau$ V

d) $10^{-4}\frac{d}{dt}\cos(100t)$ V

Sol:



KCL at node 1:

$$\frac{V_2}{V_s} = \frac{-\omega L}{R} = \frac{-100\times 10\times 10^{-3}}{10} = -\frac{1}{10}$$

$$V_2 = -\frac{V_s}{10} = -\cos 100t$$

$$\frac{V_0}{V_2} = \frac{-1/\omega C}{100} = -\frac{1}{100 \times 10 \times 10^{-6} \times 100} = -10$$

$$V_0 = -10V_2 = -10(-\cos 100t)$$

$$V_0 = 10\cos 100t$$

Current to voltage converter:

- Current to Voltage converter is an electronic circuit in which current is given as an input and voltage is produced at the output terminal. It is also known as I to V converter. Current to voltage converter is used where we need to measure output current but measuring instrument can measure voltage only.

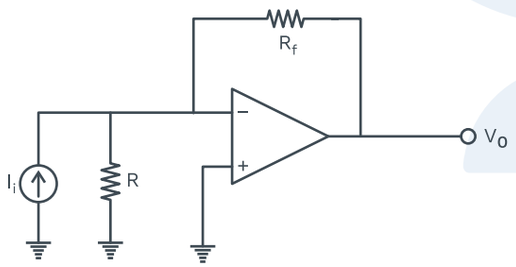
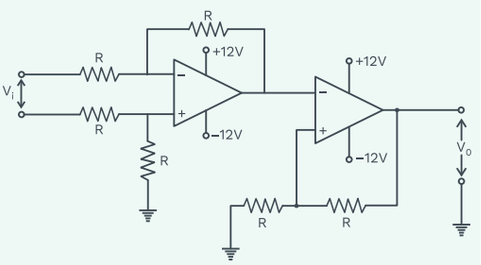


Fig. 9.20 Current to Voltage Converter

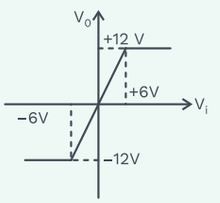
Fig. 9.20 shows a circuit of current to voltage converter. As we can see from the Fig. input current is applied at inverting terminal and a resistance (R) is connected between inverting terminal and ground whereas non-inverting terminal is grounded. Output voltage is denoted as V_0. Feedback resistor (R_f) is responsible of conversion from current to voltage therefore we can say R_f decides the characteristics curve of current to voltage converter. R_f is also known as floating because it is not connected to ground.

Previous Years' Question

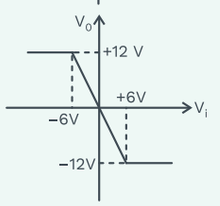
Which of the following will be characteristics of given below circuit?



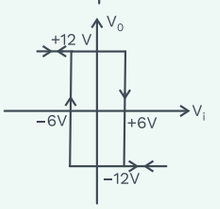
a)



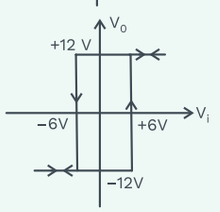
b)



c)



d)



Analysis:

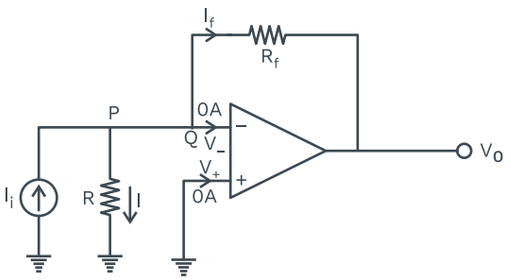


Fig. 9.21 Current to Voltage Converter

From the virtual ground concept:

$$V_- = V_+ = 0V \quad (9.42)$$

So, current through resistance R is:

$$I = \frac{(0-0)}{R} = 0A \quad (9.43)$$

Applying KCL at node Q,

$$I_i = 0 + I_f \quad (9.44)$$

$$I_i = \frac{(0-V_0)}{R_f} \quad (9.45)$$

$$V_0 = -I_i R_f \quad (9.46)$$

As we can see from equation (9.46), voltage produced at the output is proportional to the applied input current.

Voltage to current converter:

- Voltage to current converter is an electronic circuit in which voltage is given as a input and current is produced at the output terminal. It is also known as V to I converter. Voltage to current converter is used where we need to measure output voltage but measuring instrument can measure current only.
- Op-Amp based voltage to current converter can be designed if we apply input voltage at non-inverting terminal of Op-Amp. Circuit diagram of voltage to current is shown in Fig. 9.22.

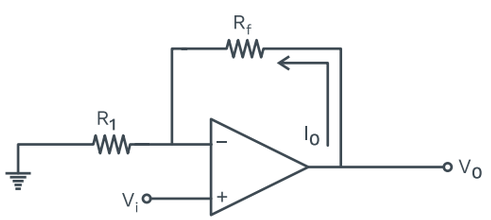


Fig. 9.22 Voltage to Current Converter

- From Fig. 9.22 we can see that input voltage is applied at non-inverting terminal of Op-Amp. Now using the concept of virtual short the voltage at non-inverting terminal and inverting will be equal. Therefore, voltage at inverting terminal will be equal to input voltage. Now applying KCL at inverting terminal, we get:

$$\frac{V_i}{R_1} - I_0 = 0 \qquad (9.47)$$

$$\Rightarrow I_0 = \frac{V_i}{R_1} \qquad (9.48)$$

Now we can see from equation (9.48) that output current (I_0) is the ratio of input voltage (V_i) to resistance (R_1).

Integrator:

Integrator is a circuit which performs integration operation. In this, output of Op-Amp will be the integral of input signal. We can form an integrator by just changing the feedback resistance with capacitor in inverting Op-Amp. Circuit diagram of integrator circuit is shown in Fig. 9.23.

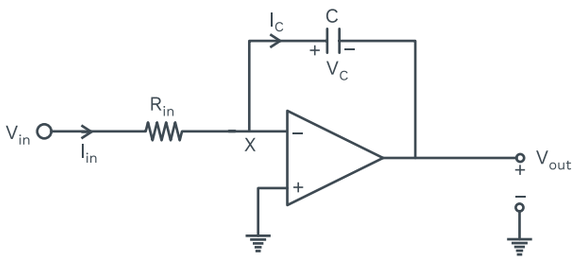


Fig. 9.23 Integrator Circuit

From virtual ground concept:

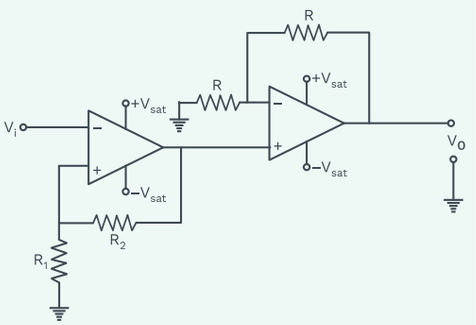
$$V_+ = V_- = V_x = 0\ V \qquad (9.49)$$

$$I_{in} = I_C \qquad (9.50)$$

$$\frac{(V_{in}-0)}{R_{in}} = C \cdot \frac{dV_C}{dt} \qquad (9.51)$$

Previous Years' Question

An operational amplifier circuits is shown in the below figure.



The output of the circuit for a given input V_i is:

a) $-\left(\frac{R_2}{R_1}\right)V_i$ **b)** $-\left(1+\frac{R_2}{R_1}\right)V_i$

c) $\left(1+\frac{R_2}{R_1}\right)V_i$ **d)** $+V_{sat}$ or $-V_{sat}$

Now, $V_C = (V_x - V_{out})$ (9.52)

$V_C = (-V_{out})$ (9.53)

From equations (9.51) and (9.53):

$$\frac{V_{in}}{R_{in}} = -C.\frac{dV_{out}}{dt} \quad (9.54)$$

$$V_{out} = \frac{-1}{R_{in}C}\int V_{in}dt \quad (9.55)$$

Practical integrator circuit:

The transfer function of the above integrator circuit will be:

$$\left|\frac{V_{out}}{V_{in}}\right| = \left|\frac{I.X_C}{I.R_{in}}\right| \quad (9.56)$$

$$\left|\frac{V_{out}}{V_{in}}\right| = \left|\frac{1}{\omega CR_{in}}\right| \quad (9.57)$$

So, in the above equation if we put $\omega = 0$, then the, gain becomes infinity,

i.e. $\left|\frac{V_{out}}{V_{in}}\right| = \infty$ (9.58)

As we can see from equation (9.58) that if V_{in} is finite then output will be infinite quantity which can't be possible therefore, to avoid this type of errors, we will add a resistance in parallel with capacitor. Now equivalent circuit is given in Fig. 9.24.

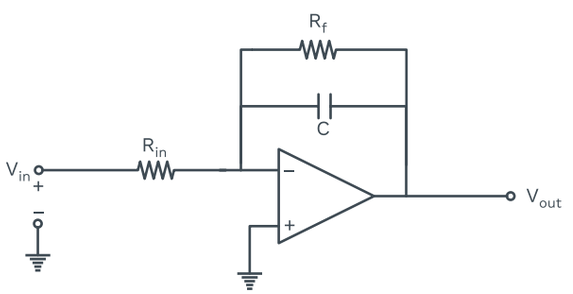


Fig. 9.24 Practical Integrator Circuit

Now, if $\omega = 0$, then

$$\left|\frac{V_{out}}{V_{in}}\right| = \left|\frac{R_f}{R_{in}}\right| \quad (9.59)$$

Differentiator:

Differentiator is a circuit which performs differentiation operation. In this output of Op-Amp will be the differential of input signal. We can form a differentiator by just changing the resistance (between inverting terminal and ground) with a capacitor in inverting Op-Amp. Circuit diagram of integrator circuit is shown in Fig. 9.25.

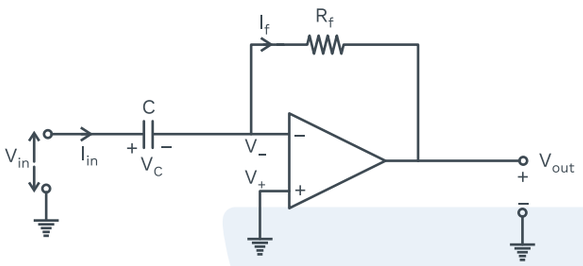


Fig. 9.25 Differentiator

Due to virtual ground concept:

$$V_+ = V_- = 0\ V \tag{9.60}$$

Now,

$$I_{in} = I_f \tag{9.61}$$

$$C.\frac{dV_C}{dt} = \frac{(V_- - V_{out})}{R_f} \tag{9.62}$$

$$C.\frac{d(V_{in} - 0)}{dt} = \frac{(0 - V_{out})}{R_f} \tag{9.63}$$

$$V_{out} = -R_f.C\frac{dV_{in}}{dt} \tag{9.64}$$

Now we can observe from equation (9.64) that output voltage (V_{out}) is multiplication of time constant ($-R_fC$) and time derivative of input voltage (V_{in}). Here minus (–) sign indicates that there is 180° phase shift between input voltage and output voltage because as we can see from Fig. 9.25 that input is connected at inverting terminal of Op-Amp.

Practical differentiator circuit:

If we find the transfer function of above differentiator circuit, then it will be:

$$\left|\frac{V_{out}}{V_{in}}\right| = \left|\frac{IR_f}{IX_C}\right|$$

$$\left|\frac{V_{out}}{V_{in}}\right| = |R_f\omega C|$$

So, in the above equation if we put $\omega = \infty$, then gain becomes infinity,

i.e. $\left|\frac{V_{out}}{V_{in}}\right| = \infty$

So, this cannot be possible to avoid this type of errors, we place a resistance in series with capacitor, and a capacitor is placed in parallel with feedback resistor. So, equivalent circuit becomes

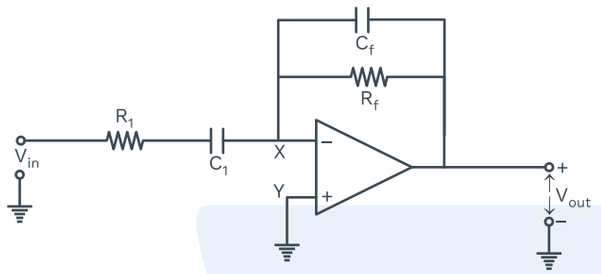


Fig. 9.26 Practical Differentiator Circuit

Now, at $\omega = \infty$ $V_{out} = 0$,

so, $\left|\frac{V_{out}}{V_{in}}\right| = 0$ (9.65)

Log amplifier:

- A log amplifier is an amplifier in which output voltage is proportional to logarithm of applied input voltage. It is also known as logarithmic amplifier. Circuit diagram of log amplifier is shown in Fig. 9.27. In this diode is in forward biased because value of input voltage is positive only.

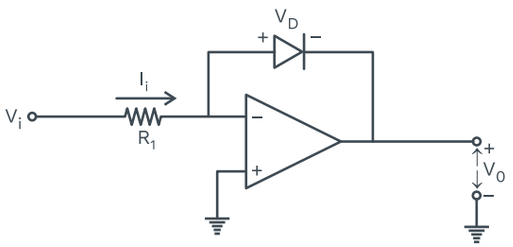


Fig. 9.27 Log Amplifier

- Diode current is:

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \quad (9.66)$$

If the diode is sufficiently forward biased then (–1) term can be neglected and diode current can be rewritten as:

$$I_D = I_S e^{\frac{V_D}{\eta V_T}} \quad (9.67)$$

The input current can be written as:

$$I_i = \frac{V_i}{R_1} \quad (9.68)$$

And the output voltage, since voltage at inverting terminal of Op-Amp which is at virtual ground, is given by:

$$V_0 = -V_D \quad (9.69)$$

And

$$I_i = I_D \quad (9.70)$$

Also we can write:

$$I_i = \frac{V_i}{R_1} = I_D = \left(I_S e^{\frac{V_D}{\eta V_T}} \right) \quad (9.71)$$

But $V_D = -V_0$ (9.72)

So, $\frac{V_i}{R_1} = I_S e^{-\frac{V_0}{\eta V_T}}$ (9.73)

Taking natural log on both sides:

$$\ln\left(\frac{V_i}{I_S R_1}\right) = -\frac{V_0}{\eta V_T} \quad (9.74)$$

$$V_0 = -\eta V_T \ln\left(\frac{V_i}{I_S R_1}\right) \quad (9.75)$$

$$V_0 = \eta V_T \ln\left(\frac{I_S R_1}{V_i}\right) \quad (9.76)$$

- A more sophisticated circuit uses bipolar transistor to eliminate I_S in log term. This circuit is shown in Fig. 9.28:

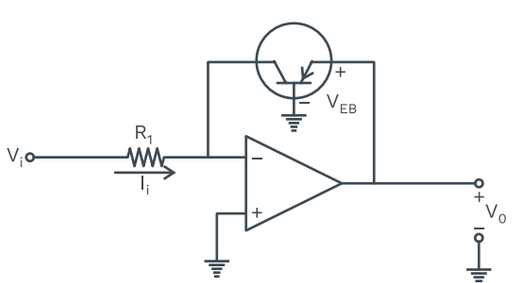


Fig. 9.28 Log Amplifier

For the above configuration, output voltage is given as:

$$V_0 = \eta V_T \ln\left(\frac{R_1 I_{co}}{V_i}\right) \qquad (9.77)$$

Antilog or exponential amplifier:

- An anti-log amplifier is an amplifier in which output voltage is proportional to exponential of applied input voltage. It is also known as exponential amplifier. Circuit diagram of antilog amplifier is shown in Fig. 9.29. In this, diode is in forward biased because value of input voltage is positive only.

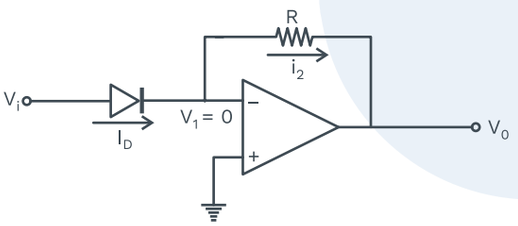


Fig. 9.29 Antilog or Exponential Amplifier

Since V_1 is at a virtual ground,
We can write for $V_i > 0$,

$$I_D = I_S e^{\frac{V_i}{\eta V_T}} \qquad (9.78)$$

And $V_0 = -i_2 R = -I_D R$ (9.79)

$$V_0 = -I_S R\, e^{\frac{V_i}{\eta V_T}} \qquad (9.80)$$

- A more sophisticated circuit using BJT is shown in Fig. 9.30.

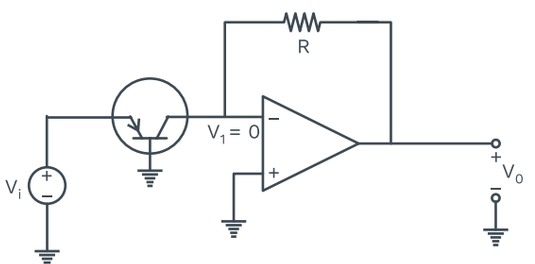
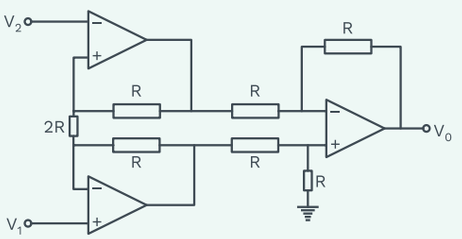


Fig. 9.30 Antilog or Exponential Amplifier

Previous Years' Question

Given that the Op-Amps in the Fig. are ideal, the output voltage V_0 is:



a) $(V_1 - V_2)$ **b)** $2(V_1 - V_2)$

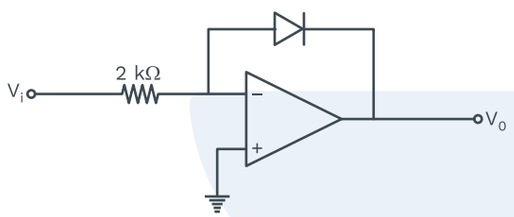
c) $\frac{(V_1 - V_2)}{2}$ **d)** $(V_1 + V_2)$

Output voltage for the above configurations is given as:

$$V_0 = -RI_{CO}e^{\frac{V_i}{\eta V_T}} \qquad (9.81)$$

Example 18:

Consider the circuit shown below. Value of diode current is given by $I = I_S$ exp (V/V_T), where V = voltage across diode. If $V_i = 2$ V, then $V_0 = V_{01}$, whereas if $V_i = 4$ V then $V_0 = V_{02}$. Find the relationship between V_{02} and V_{01}.



a) $V_{02} = \sqrt{2}\, V_{01}$ **b)** $V_{02} = e^2\, V_{01}$

c) $V_{02} = V_{01} \ln 2$ **d)** $V_{01} - V_{02} = V_T \ln 2$

Sol:

Lets inverting terminal is represented by A and non-inverting terminal represented by B.

$V_A = V_B = 0$ [virtual short]

Case 1. When $V_i = 2$ V, $V_0 = V_{01}$

$$I_1 = \frac{(V_i - 0)}{2\,k\Omega} = \frac{V_i}{2}\,mA$$

$$\Rightarrow I_1 = \frac{2}{2}\,mA = 1\ mA$$

V_0 will be negative as the Op-Amp is in inverting mode.

Now, $I_1 = I_{D1}$

$$I_1 = I_S \exp\left(-\frac{V_{01}}{V_T}\right) \qquad (I)$$

Or, $1 = I_S \exp\left(\frac{-V_{01}}{V_T}\right)$

Case 2. When $V_i = 4$ V, $V_0 = V_{02}$

$$I_2 = \frac{(V_i - 0)}{2k\Omega} = \frac{V_i}{2}mA$$

$$\Rightarrow I_2 = \frac{4}{2}\text{mA} = 2\text{mA}$$

$$I_2 = I_{D2}$$

$$I_2 = I_s \exp\left(\frac{-V_{02}}{V_T}\right) \qquad \text{(II)}$$

$$2 = I_S \exp\left(\frac{-V_{02}}{V_T}\right)$$

From equations (1) and (2),

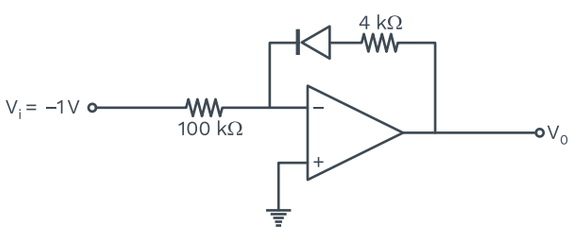
$$\frac{2}{1} = \frac{\exp\left(\frac{-V_{02}}{V_T}\right)}{\exp\left(\frac{-V_{01}}{V_T}\right)}$$

$$2 = \exp\left(\frac{-V_{02} + V_{01}}{V_T}\right)$$

$$V_{01} - V_{02} = V_T \ln 2$$

Example 19:

The below given circuit contains an Op-Amp and a diode. Relation between current and voltage across diode is given by $I = I_0\left(e^{V/V_T} - 1\right)$. Where V = Voltage across diode, I_0 = 1 μA and V_T = 25 mV. What will the value of output voltage if input voltage is –1 V (V_i = –1 V)?



a) 0 V **b)** 0.1 V **c)** 0.7 V **d)** 1.1 V

Sol:

Given Op-Amp is an inverting amplifier and input is negative, so the output would be positive. Which implies that diode is forward biased.

Now, let inverting terminal is represented by A and non-inverting terminal by B.

$V_A = V_B = 0$ [virtual short]

$I_1 = I_D$

$$\frac{0-(-1)}{100\times10^{3}} = I_0\left(e^{\frac{V}{V_T}}-1\right)$$

$$\frac{1}{10^{5}} = \frac{1}{10^{6}}\left(e^{\frac{V}{25\ mV}}-1\right)$$

$$11 = e^{V/25\,mV}$$

$$V = 25\ \ln 11\ mV$$

$$V = 0.06\ V$$

Now, $\frac{(V_0 - V)}{4k} = \frac{1}{100k}$

$$V_0 = 0.1\ V$$

Differential amplifier:

- An ideal differential amplifier can be defined as an amplifier which amplify only the difference of two input signal and rejects if any common signal is applied at two input terminal.
- Op-Amp based differential amplifier circuit is shown in Fig. 9.31. In this V_{i_1} is applied at inverting terminal whereas V_{i_2} is applied at non-inverting. For analysing circuit, we will use concept of virtual-short and super-position.

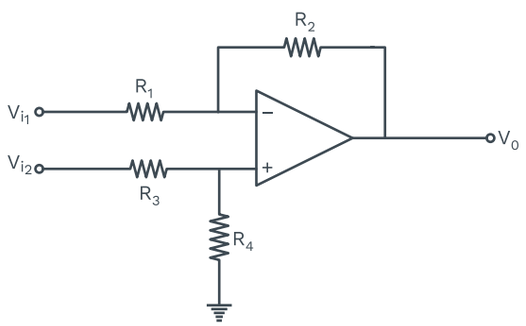
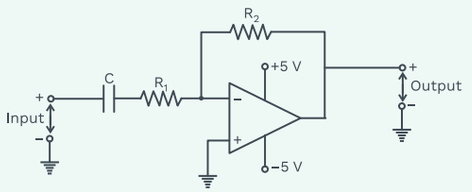


Fig. 9.31 Op-Amp as a Differential Amplifier

Previous Years' Question

The circuit shown is a:



a) Low-pass filter with $f_{3dB} = \frac{1}{(R_1 + R_2)C}$ rad/s

b) High-pass filter with $f_{3dB} = \frac{1}{R_1 C}$ rad/s

c) Low-pass filter with $f_{3dB} = \frac{1}{R_1 C}$ rad/s

d) High-pass filter with $f_{3dB} = \frac{1}{(R_1 + R_2)C}$ rad/s

From Fig. 9.32 first we will make $V_{i_2} = 0$ and due to this V_{2a} is equal to zero therefore current through resistance (R_3) and resistance (R_4) is zero. Now resulting circuit will behave as inverting amplifier. Now output voltage is given as:

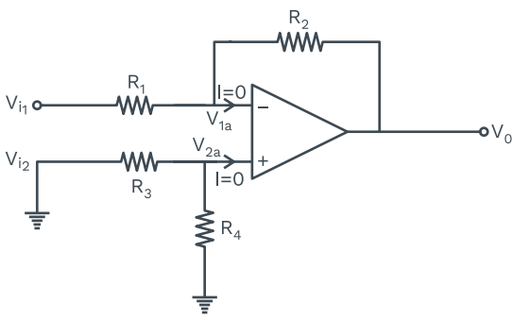


Fig. 9.32 Difference Amplifier with $V_{i_2} = 0$

$$V_{01} = -\frac{R_2}{R_1}.V_{i_1} \qquad (9.82)$$

From Fig. 9.33 now we will make $V_{i_1} = 0$ and since current through the Op-Amp is zero. Now resulting circuit will behave as non-inverting amplifier. Now with the help of virtual-short and super-position we will find the output voltage.

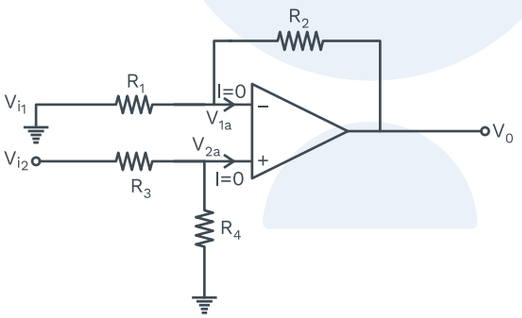


Fig. 9.33 Difference Amplifier with $V_{i_1} = 0$

$$V_{2b} = \left(\frac{R_4}{R_3 + R_4}\right) \cdot V_{i_2} \qquad (9.83)$$

As we know that from the concept of virtual-short, we will get $V_{2a} = V_{2b}$. Now circuit behaves as a non-inverting amplifier. Output voltage is given by:

$$V_{02} = \left(1 + \frac{R_2}{R_1}\right) V_{2b} \qquad (9.84)$$

Substituting equation (9.83) into (9.84), we get:

$$V_{02} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) \cdot V_{i_2} \qquad (9.85)$$

$$\text{Or, } V_{02} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4 / R_3}{1 + R_4 / R_3}\right) \cdot V_{i_2} \qquad (9.86)$$

Since, the net output voltage is the sum of individual terms, we have:

$$V_0 = V_{01} + V_{02} \quad (9.87)$$

$$V_0 = \left(1+\frac{R_2}{R_1}\right)\left(\frac{R_4/R_3}{1+R_4/R_3}\right)V_{i2} - \left(\frac{R_2}{R_1}\right)V_{i_1} \quad (9.88)$$

A property of ideal differential amplifier is that the output voltage is zero when $V_{i_1} = V_{i_2}$, on analysis of equation (9.88) this condition is met if:

$$\frac{R_4}{R_3} = \frac{R_2}{R_1} \quad (9.89)$$

The output voltage will be:

$$V_0 = \frac{R_2}{R_1}\left(V_{i_2} - V_{i_1}\right) \quad (9.90)$$

which indicates that, this amplifier has a differential gain of $A_d = \frac{R_2}{R_1}$

Calculation of differential input resistance:

Differential input resistance can be calculated by using a test voltage (V_i) and test current (I_i). Ratio of test voltage and test current is differential input resistance. Now calculation of differential input resistance can be done by the circuit shown in Fig. 9.34. We have assumed here that $R_1 = R_3$ and $R_2 = R_4$. Now input resistance is given by:

$$R_i = \frac{V_i}{I_i}$$

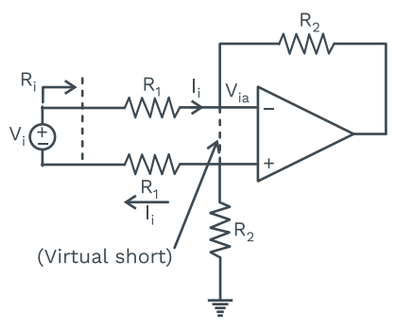


Fig. 9.34 Differential Amplifier

Rack Your Brain

Why does Op-Amp have high CMRR?

By virtual short concepts, we can write the following loop equations:

$$V_i = I_iR_1 + I_iR_1 \quad (9.91)$$

$$V_i = I_i(2R_1) \quad (9.92)$$

$$\frac{V_i}{I_i} = 2R_1 \quad (9.93)$$

Therefore, the input resistance is:

$$R_i = 2R_1 \quad (9.94)$$

Differential amplifier gain:

Differential amplifier gain is defined as the ratio of output voltage and difference of input applied voltage.

- From the previous calculation, we have the output voltage V_{out} as:

$$V_{out} = \frac{R_2}{R_1} \cdot (V_1 - V_2) \quad (9.95)$$

So, differential amplifier gain is given by:

$$A_d = \left(\frac{V_{out}}{V_1 - V_2}\right) = \frac{R_2}{R_1} \quad (9.96)$$

Common mode gain:

- For an ideal amplifier, output voltage (V_0) is zero if $V_{i_1} = V_{i_2}$. But if we see equation (9.88) we will observe that output voltage is not zero if $\frac{R_4}{R_3} \neq \frac{R_2}{R_1}$
- In common mode input, value of input voltage is equal that is $V_{i_1} = V_{i_2}$. For a common mode, input voltage is defined as:

$$V_{cm} = \frac{(v_{i_1} + v_{i_2})}{2} \quad (9.97)$$

- The common mode gain is defined as:

$$A_{cm} = \frac{V_0}{V_{cm}} \quad (9.98)$$

Common mode rejection ratio:

Common mode rejection ratio is defined as modulus of ratio of differential mode gain to common mode gain. For a differential amplifier Fig. of merit is common mode rejection ratio. It is denoted by CMRR.

$$CMRR = \left|\frac{A_d}{A_{cm}}\right| \quad (9.99)$$

Usually, the CMRR is expressed in decibels, as follows:

$$CMRR(dB) = 20 \log_{10} \left|\frac{A_d}{A_{cm}}\right| \quad (9.100)$$

- If we take ideal condition, then value of common mode rejection ratio is infinite for a differential amplifier, but in practical case value should be as large as possible.
- Common mode rejection ratio (CMRR) is one of the most important specifications of differential amplifier because CMRR indicates the rejection of common-mode signal at input terminal which eventually decides the ability to minimize the noise in communication design.

Output voltage in differential amplifier:

$$V_0 = V_{ocm} + V_{od} \quad (9.101)$$

$$V_0 = A_{cm}\, V_{cm} + A_d\, V_d \quad (9.102)$$

$$V_0 = A_d V_d \left(1 + \frac{A_{cm} V_{cm}}{A_d V_d} \right) \quad (9.103)$$

$$V_0 = A_d V_d \left(1 + \frac{V_{cm}}{V_d \text{CMRR}} \right) \quad (9.104)$$

Where,

V_{ocm} = Output due to common mode input.

V_{od} = Output due to differential mode input.

A_{cm} = Common mode gain.

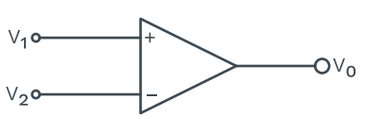
A_d = Differential mode gain.

V_{cm} = Common mode input = $\frac{\left(V_{i_1} + V_{i_2}\right)}{2} = \frac{\left(V_+ + V_-\right)}{2}$

V_d = Differential mode input = $\left(V_{i_1} - V_{i_2}\right) = \left(V_+ - V_-\right)$

Example 20:

A differential amplifier is given below. Input voltage at non-inverting terminal is 0.55 V and voltage at inverting terminal is 0.45 V. What will be the value of output voltage of differential amplifier if CMRR is 40 dB and differential mode gain is 100?



a) 10 V **b)** 10.5 V **c)** 11 V **d)** 15 V

Sol:

$$\text{CMRR} = 20 \log_{10} \frac{A_d}{A_{cm}}$$

$$40=20\log_{10}\left(\frac{A_d}{A_{cm}}\right)$$

$$\frac{A_d}{A_{cm}}=100$$

$$\therefore A_{cm}=\frac{A_d}{100}=\frac{100}{100}=1$$

$$V_0=A_d\left(V_1-V_2\right)+A_{cm}\left(\frac{V_1+V_2}{2}\right)$$

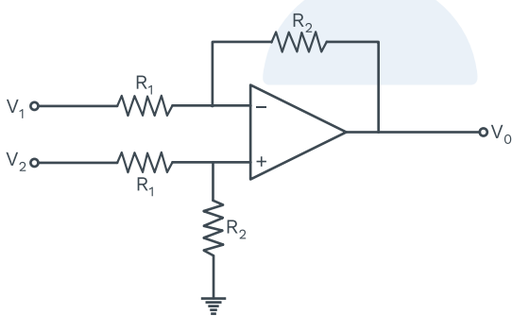
$$V_0 = 100\left(0.55-0.45\right)+1\times\left(\frac{0.55+0.45}{2}\right)$$

$$V_0 = 10+0.5$$

$$V_0 = 10.5\ V$$

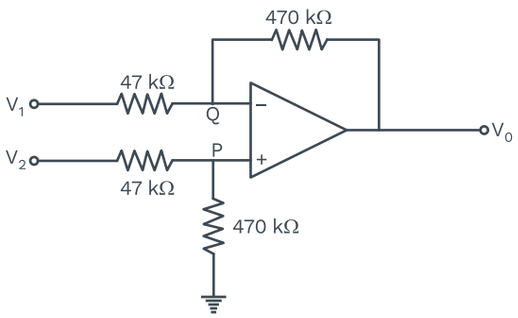
Example 21:

A differential amplifier is given below, which is constructed by using an Op-Amp. Find the value of input resistance by looking the terminals V_1 and V_2 with respect to ground if value of R_1 = 47 kΩ and R_2 = 470 kΩ?



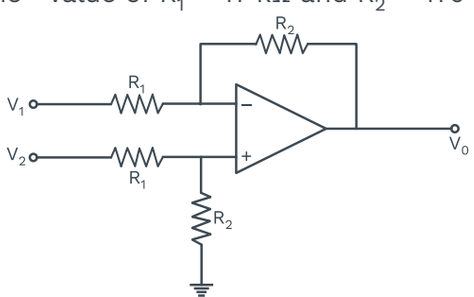
a) 517 kΩ for V_1 and 47 kΩ for V_2
b) 517 kΩ for V_1 and 517 kΩ for V_2
c) 47 kΩ for V_1 and 517 kΩ for V_2
d) 47 kΩ for V_1 and 47 kΩ for V_2

Sol:



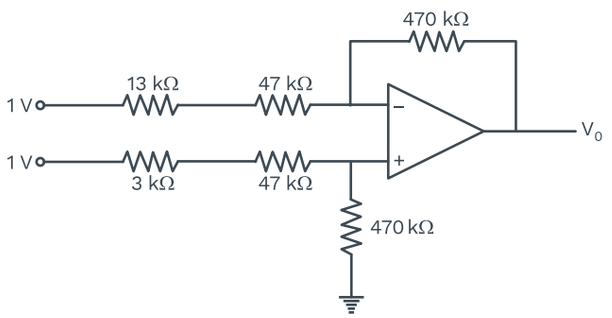
Apply superposition theorem put $V_2 = 0$,
So, voltage at:
$P = 0$
By virtual ground concept voltage at $Q = 0$
So, input impedance seen from V_1 with respect to ground is:
$R_1 = 47\ k\Omega$
Now, put $V_1 = 0$.
The input impedance from V_2 with respect to ground = $R_1 + R_2$
$= (47 + 470)\ k\Omega$
$= 517\ k\Omega$

Example 22:
A differential amplifier is given below, which is constructed by using an Op-Amp. Find the value of output voltage if voltage source is connected at both terminal and voltage at the terminals V_1 and terminal V_2 is +1 V with respect to ground, whereas internal resistance of source is 13 kΩ and 3 kΩ respectively, assume value of $R_1 = 47\ k\Omega$ and $R_2 = 470\ k\Omega$.



a) 0 V **b)** 0.15 V **c)** 1.5 V **d)** 10 V

Sol:
Effective circuit will be:



From equation (9.88):

$$V_0 = 1\left(\frac{470}{470+50}\right)\left[1+\frac{470}{60}\right]-1\left[\frac{470}{60}\right]$$

$$V_0 = 0.15 \text{ V}$$

Instrumentation amplifier:

- An instrumentation amplifier is a differential amplifier which is used to amplify very low-level signal. An instrumentation amplifier eliminates the need of input impedance matching, therefore it is suitable for use in measurement and test equipment. An instrumentation amplifier is fitted with input buffer amplifier. The main function of instrumentation amplifier is that it is used for amplification of difference in the input signal.
- Some of the important features of instrumentation amplifier are:
  1) High CMRR
  2) High gain accuracy
  3) High gain stability with very low temperature coefficient
  4) Low output impedance
  5) Low DC offset

Instrumentation amplifier circuit design:
Circuit design of instrumentation amplifier using 3 Op-Amp is shown in Fig. 9.35. By using 3 Op-Amp in an instrumentation amplifier input impedance becomes very high which helps in measurement of signal from transducer.

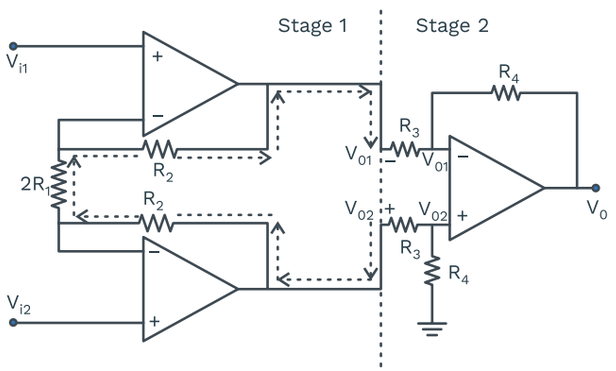


Fig. 9.35 Three Op-Amp Circuit of an Instrumentation Amplifier.

- First stage of instrumentation amplifier is voltage follower circuit whereas second stage is difference amplifier. In first stage there are two buffer amplifiers having very high input impedance, which makes them instrumentation amplifier suitable for measurement.

Now, we will derive the condition for stage 1:
At the input of voltage follower circuit, current equation can be written as:

$$I = \frac{V_{i_2} - V_{i_1}}{2R_1} \quad (9.105)$$

Let $V_{i_2} - V_{i_1} = V_{id}$ (9.106)

Where, V_{id} = Input difference voltage.
Applying KVL at the first stage of the above circuit:

$$IR_2 + I(2R_1) + IR_2 = V_{02} - V_{01} \quad (9.107)$$

$$2IR_2 + 2IR_1 = V_{02} - V_{01} \quad (9.108)$$

Substituting the value of I in the above equation:

$$\frac{2V_{id}}{2R_1} \cdot R_2 + \frac{2V_{id}}{2R_1} \cdot R_1 = V_{02} - V_{01} \quad (9.109)$$

on simplifying we get:

$$\frac{V_{id}}{R_1} \cdot R_2 + \frac{V_{id}}{R_1} \cdot R_1 = V_{02} - V_{01} \quad (9.110)$$

Taking like term as common, we get:

$$V_{id}\left(\frac{R_2}{R_1} + 1\right) = V_{02} - V_{01} \quad (9.111)$$

The output of the above-represented circuit is V_0, which can be written as:

$$V_0 = (V_{02} - V_{01}) \cdot \frac{R_4}{R_3} \qquad (9.112)$$

From equations (9.112) and (9.111):

$$V_0 = V_{id}\left(\frac{R_2}{R_1} + 1\right)\left(\frac{R_4}{R_3}\right) \qquad (9.113)$$

So, the above equation is for overall output voltage of an instrumentation amplifier.

Some of the advantage of instrumentation amplifier:

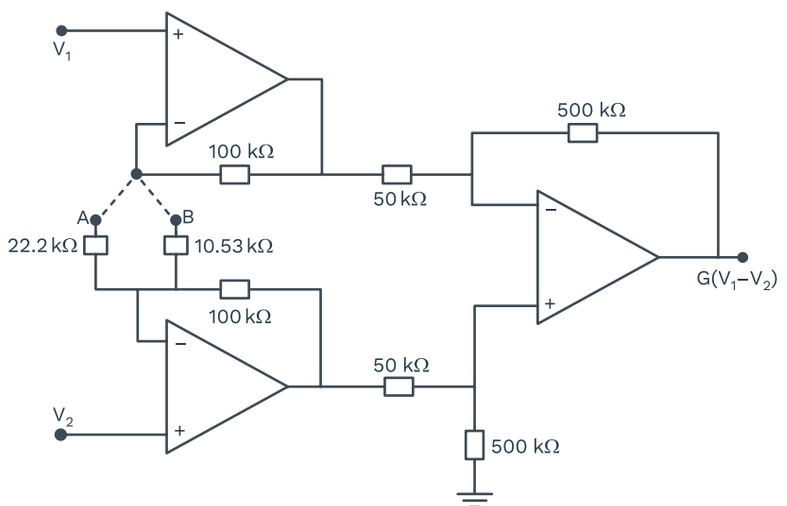
1) Instrumentation amplifier has high input impedance.
2) Instrumentation amplifier has low output impedance.
3) Instrumentation amplifier has high value of CMRR.
4) Instrumentation amplifier consumes less power.

Disadvantage of instrumentation amplifier:

- As we know that instrumentation amplifier is used for amplification of low-level signal which is transmitted for long distance. There is some noise at the time of amplification of original signal. which is also amplified, therefore signal get distorted.

Example 23:

The below given Fig. is an instrumentation amplifier. Now switch is changed from A to B. What will be value of gain (G) before and after changing the switch.



a) 45, 95 **b)** 50, 100 **c)** 100, 200 **d)** 90, 180

Sol:

The gain of the instrumentation amplifier is given as:

$$G = \frac{V_{out}}{V_2 - V_1}$$

$$G = \left(1 + \frac{2R_1}{R_{gain}}\right)\frac{R_3}{R_2}$$

At position 'A' the gain is:

$R_1 = 100\ k\Omega$

Here, $R_{gain} = 22.2\ k\Omega$

$R_2 = 50\ k\Omega$

$$G = \left(1 + \frac{2 \times 100}{22.2\,\Omega}\right) \times \frac{500}{50}$$

$G = 100$

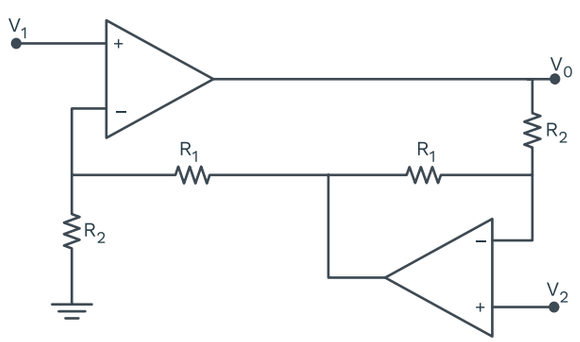
At position 'B' the $R_{gain} = 10.53\ k\Omega$, now gain is:

$$G = \left(1 + \frac{2 \times 100}{10.53}\right) \times \frac{500}{50}$$

$G = 200$

Example 24:

An instrumentation amplifier is given below. Find the value of output voltage in terms of V_1, V_2, R_1 and R_2?



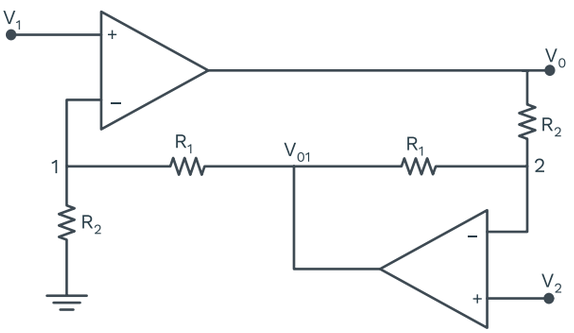
a) $V_0 = \left(1 + \frac{R_2}{R_1}\right)(V_2 - V_1)$

b) $V_0 = \left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1)$

c) $V_0 = \left(\frac{2R_2}{R_1}\right)(V_2 - V_1)$

d) $V_0 = \left(1 + \frac{2R_1}{R_2}\right)(V_2 - V_1)$

Sol:



Voltage at node 1 = V_1
(Virtual ground concept)

and $V_1 = \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{01}$

or, $V_{01} = \left(1 + \frac{R_1}{R_2}\right) \cdot V_1$

Voltage at node 2 = V_2
Writing KCL at node 2 we have:

$$\frac{V_0 - V_2}{R_2} = \frac{V_2 - V_{01}}{R_1}$$

$$\frac{V_0}{R_2} = V_2\left[\frac{1}{R_2} + \frac{1}{R_1}\right] - \frac{1}{R_1}\left[1 + \frac{R_1}{R_2}\right] \cdot V_1$$

$$V_0 = V_2\left[\frac{R_1 + R_2}{R_1}\right] - \frac{R_2}{R_1}\left[1 + \frac{R_1}{R_2}\right] \cdot V_1$$

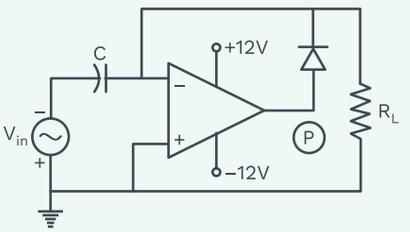
$$V_0 = (V_2 - V_1)\left(1 + \frac{R_2}{R_1}\right)$$

Comparators:

- A comparator is an electronic device, which produce a constant output by comparing two input which are applied at input terminal.
- Output of comparator decides which input is greater. For an Op-Amp based comparator it indicates whether inverting terminal is greater or non-inverting terminal is greater.

Previous Years' Question

For a given sinusoidal input voltage, the voltage waveform at point P of the clamper circuit shown in figure will be:





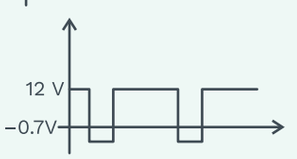
a)



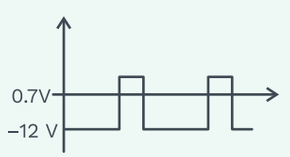
b)

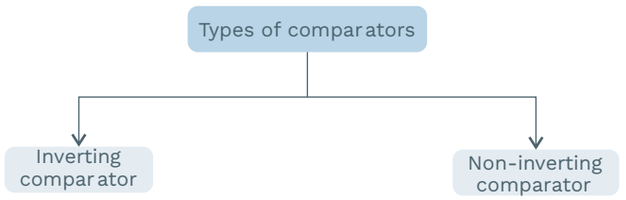


c)



d)





Inverting comparator:

- An inverting comparator is an Op-Amp based comparator in which input voltage is applied at inverting terminal, whereas reference voltage is applied at non-inverting terminal and as it is a comparator circuit, the input applied at inverting terminal is compared to non-inverting terminal of Op-Amp.

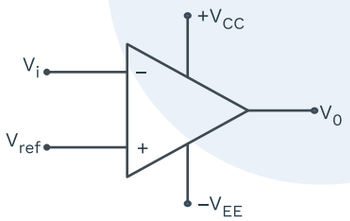


Fig. 9.36 Circuit Diagram of Inverting Comparator

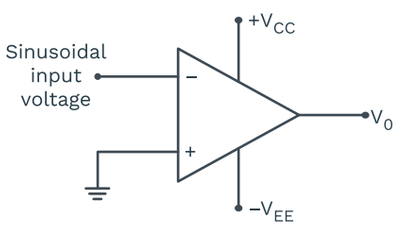
- As the operation of inverting comparator is very simple, output of inverting comparator will be either $+V_{sat}$ or $-V_{sat}$ depending on the applied input voltage and reference voltage which is applied at non-inverting terminal.
- Output of inverting amplifier will be $+V_{sat}$ if applied voltage is less than reference voltage whereas output will be $-V_{sat}$ if applied voltage is greater than reference voltage.

Grey Matter Alert!

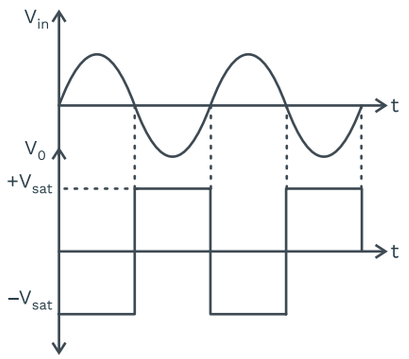
In an Op-Amp based comparator there is no feedback loop present in the amplifier.

Example 25:

A sinusoidal voltage is applied at input terminal of inverting comparator and reference voltage is zero volts. Draw the output waveform for inverting comparator.



Sol:



Non-inverting comparator:

- A non-inverting comparator is an Op-Amp based comparator in which input voltage is applied at non-inverting terminal whereas reference voltage is applied at inverting terminal as it is a comparator circuit, the input applied at non-inverting terminal will be compared to inverting terminal of Op-Amp.

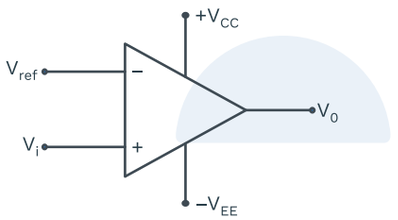
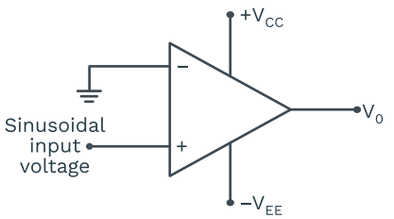


Fig. 9.37 Non-inverting Comparator

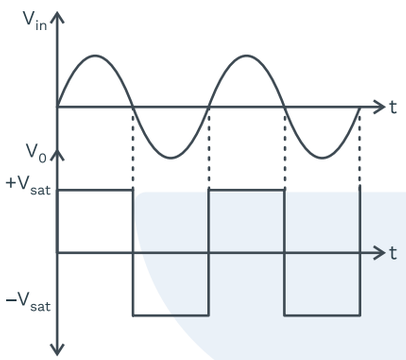
- As the operation of non-inverting comparator is very simple, output of non-inverting comparator will be either $+V_{sat}$ or $-V_{sat}$ depending on the applied input voltage and reference voltage which is applied at inverting terminal.
- Output of inverting amplifier will be $+V_{sat}$, if applied voltage is greater than reference voltage while output will be $-V_{sat}$ if applied voltage is less than reference voltage.

Example 26:

A sinusoidal voltage is applied at input terminal of non-inverting comparator and reference voltage is zero volts. Draw the output waveform for non-inverting comparator.

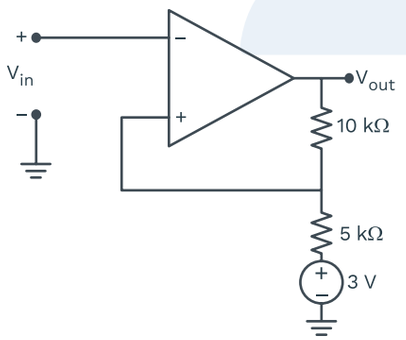


Sol:



Example 27:

For the given Op-Amp, find the value of upper threshold voltage and lower threshold voltage of amplifier. Given output saturation voltage is ±15 V.



a) +5 V and −5 V
b) +7 V and −3 V
c) −3 V and +7 V
d) +3 V and −3 V

Sol:

For the given circuit the upper and lower threshold voltage are given by:

$$\text{UTP} = \frac{R_2}{R_1 + R_2} \cdot V_{sat} + \frac{R_1}{R_1 + R_2} \cdot V_\gamma \quad \text{(upper threshold voltage)}$$

$$\text{and LTP} = -\frac{R_2}{R_1 + R_2} \cdot V_{sat} + \frac{R_1}{R_1 + R_2} \cdot V_\gamma \quad \text{(lower threshold voltage)}$$

here, $R_1 = 10\ k\Omega$, $R_2 = 5\ k\Omega$, $V_\gamma = 3\ V$

and $V_{sat} = 15\ V$

$$\therefore UTP = \frac{5}{5+10} \times 15 + \frac{10}{5+10} 3$$

$$= 5 + 2 = 7\ V$$

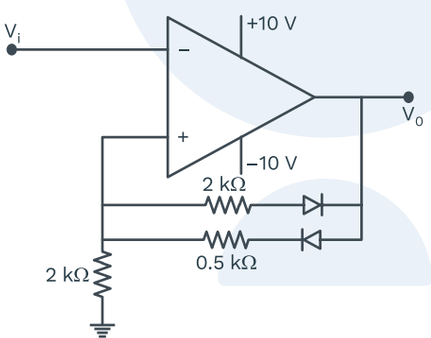
$$LTP = \frac{-5}{5+10} \times 15 + \frac{10}{5+10} \times 3$$

$$= -5 + 2$$

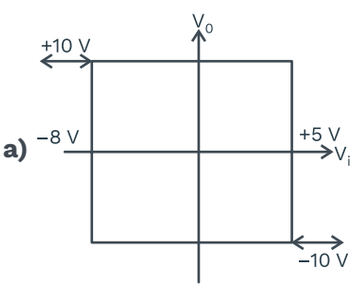
$$= -3\ V$$

Example 28:

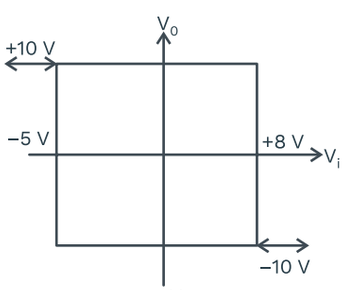
An ideal operation amplifier is shown in the below Fig.. Which among the following depicts the correct transfer characteristics curve for Op-Amp? (Assume diodes are ideal and cut-in voltage is zero.)



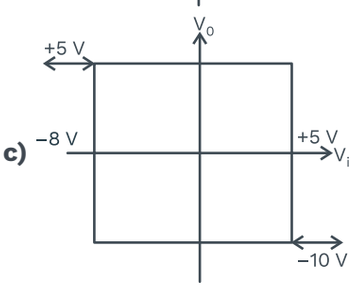
a)



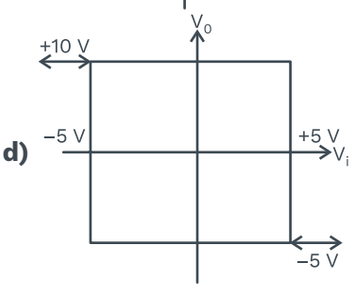
b)



c)



d)



Sol:

The circuit will behave as vibrator.

When V_0 = +10 V, lower diode will be forward biased. Therefore,

$V_i = V_{UT}$

$V_{UT} \rightarrow$ upper threshold voltage

$$V_i = \frac{10 \times 2}{0.5 + 2}$$

V_i = 8 V

When V_0 = -10 V

Upper diode will be forward biased

$V_i = V_{LT}$

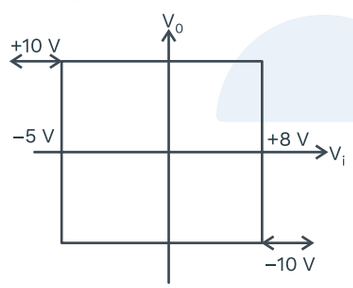
$V_{LT} \rightarrow$ Lower threshold voltage

$$V_{LT} = \frac{V_0(2k)}{2k + 2k}$$

$$V_{LT} = \frac{V_0}{2} = -\frac{10}{2}$$

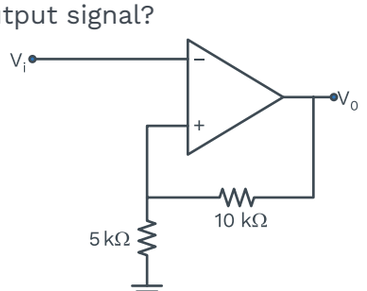
$V_{LT} = -5$ V

Hence transfer characteristics



Example 29:

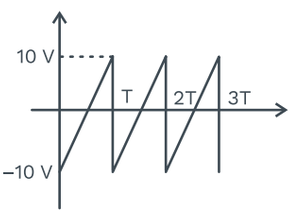
A symmetrical sawtooth wave (20 $V_{P\text{-}P}$) having positive slope and average value zero is applied at inverting terminal in circuit shown below. Assuming Op-Amp is ideal with peak-to-peak symmetrical swing of 30 V. What will be the average value of output signal?



a) 5V **b)** 10V **c)** −5V **d)** 7.5V

Sol:

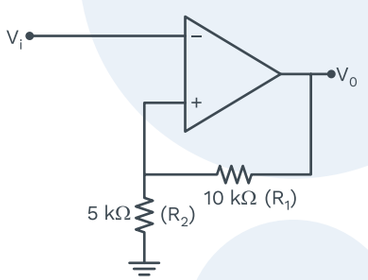
Given, sawtooth wave is



so equation for one period

$$y = \frac{20}{T}t - 10 \text{ (for first period)}$$

For the circuit given



Since output is symmetrical and peak to peak = 30 V

So,

$$+V_{sat} = 15 \text{ V}$$

$$-V_{sat} = -15 \text{ V}$$

Now, upper threshold voltage

$$V_{UT} = \frac{R_2}{R_1 + R_2} \times \left(+V_{sat}\right)$$

$$= \frac{5}{15} \times 15 = 5 \text{ V}$$

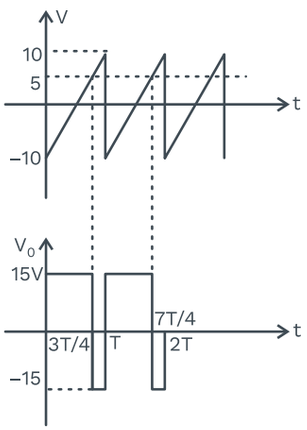
Lower threshold voltage

$$V_{LT} = \frac{R_2}{R_2 + R_1} \times \left(-V_{sat}\right) = -5 \text{ V}$$

so time at which input will be +5 V

$$5 = \frac{20}{T}t - 10 \rightarrow t = \frac{3}{4}T$$

Now output waveform



So average value of output

$$V_{avg} = \frac{15 \times 3T / 4 + (-15) \times T / 4}{T}$$

$$= \frac{45}{4} - \frac{15}{4} = \frac{30}{4} = 7.5 \text{ V}$$

Precision diodes:

Ordinary diodes can not be used for rectification purpose if operating voltage range is below the cut-in-voltage of the diode. A circuit will act as ideal diode or signal-processing rectifier which can be used for rectification of voltage below the cut-in-voltage of the diode and can be designed if a diode is placed in the feedback loop of Op-Amp.

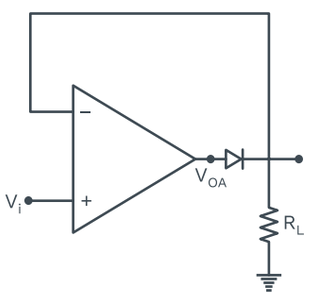


Fig. 9.38 Precision Rectifier

- Arrangement of precision diode is shown in Fig. 9.38. In this, a single diode is arranged such that it acts as a non-inverting precision half-wave rectifier circuit.

- If input voltage (V_i) applied is positive then output voltage (V_{OA}) will become positive and diode becomes forward biased and closed loop condition is achieved for Op-Amp hence the output voltage will be equal to input voltage ($V_0 = V_i$). Now, If input voltage (V_i) applied is negative then output voltage (V_{OA}) will become negative and diode becomes reverse biased and closed loop condition will break for Op-Amp, therefore output voltage will be equal to zero ($V_0 = 0$). Therefore, Output waveform will look like:

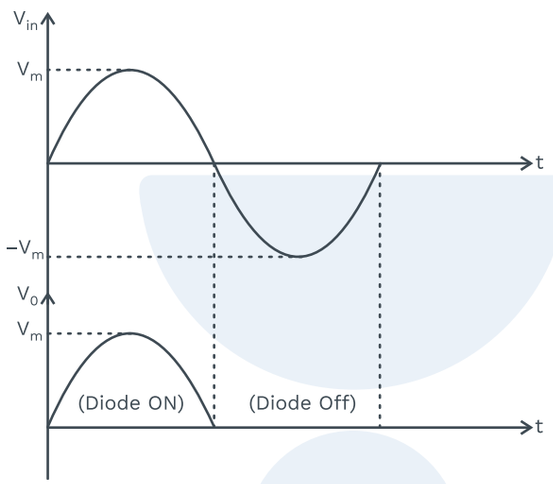


Fig. 9.39 Waveforms of Precision Rectifier

The above circuit is an example of non-linear circuit but still we achieve some linear portion at output, if input voltage is less than zero, then output for negative input voltage swings to negative saturation level. So, we can say that above circuit is saturation form of circuit, therefore frequency response is also limited.

Application of precision diodes:

Following are some of the applications of precision diode:

1) Half-wave rectifier
2) Full-wave rectifier
3) Peak value detector
4) Clipper and clamper circuit

1) Half-wave rectifier:

An inverting amplifier can act as half-wave rectifier, if we add two diodes as shown in Fig. 9.40. If we apply positive voltage, then diode D_1 starts conducting therefore output voltage V_{OA} will become negative (–0.7 V) due to voltage drop across diode D_1. Now due to negative voltage across output node, diode D_2 will be in reverse bias therefore there will not be any current across feedback resistance (R_f) and all current flows through D_1, finally output voltage V_0 will become zero.

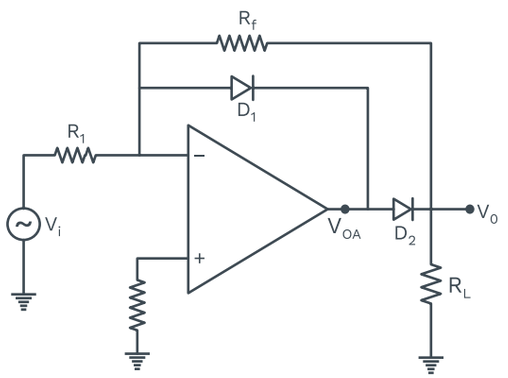


Fig. 9. 40 Ideal Half Wave Rectifier

Now if we apply negative input voltage, then diode D_1 will become off and diode D_2 will conduct. Due to negative voltage applied at input, output of Op-Amp V_{OA} will becomes positive and diode D_2 starts conducting. Now there will be current through feedback resistance and circuit will act as an inverter. Now if $R_f = R_1$ then output voltage V_0 will be positive. For half-wave rectifier, input-output waveform is shown in Fig. 9.41.

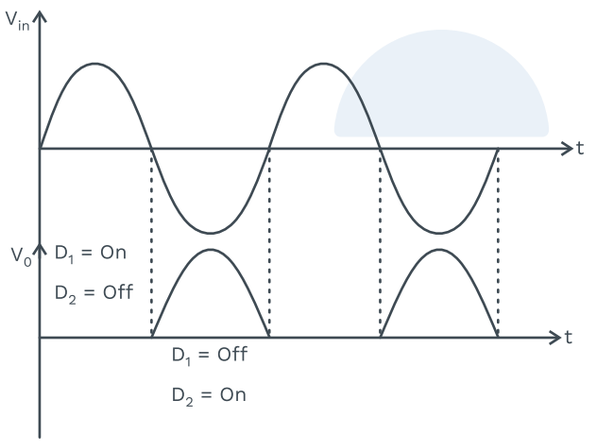


Fig. 9.41 Half Wave Rectifier Waveforms

The circuit operation can mathematically be expressed as,

$V_0 = 0$ When $V_i > 0$ and (9.114)

$$V_0 = \frac{-R_f}{R_1} V_i \quad \text{for} \quad V_i < 0 \tag{9.115}$$

The voltage V_{OA} at the Op-Amp output is

$$V_{OA} = -0.7 \text{ V for } V_i > 0 \quad (9.116)$$

$$V_{OA} = \frac{R_f}{R_1} V_i + 0.7 \text{ V} \quad \text{for } V_i < 0 \quad (9.117)$$

Advantage of precisions half-wave rectifier:

- It is a non-saturation amplifier.
- The inverting characteristics of the output voltage V_0 can be circumvented by the use of an additional inversion for achieving a positive output.

2) Full wave rectifier:

A full wave rectifier circuit is shown in Fig. 9.42 using two Op-Amp and two diodes.

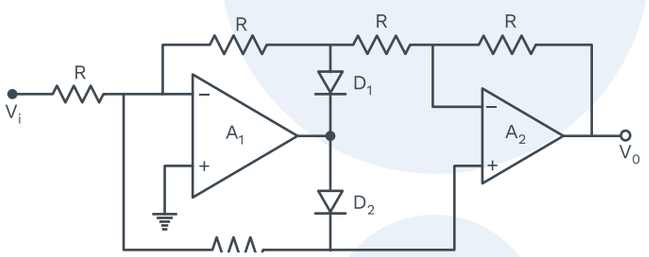


Fig. 9.42 Full Wave Rectifier Circuit

If we apply positive input voltage, then diode D_1 will conduct and diode D_2 will be off. Both the Op-Amp A_1 and Op-Amp A_2 will act as an inverter as we can see from Fig. 9.43.

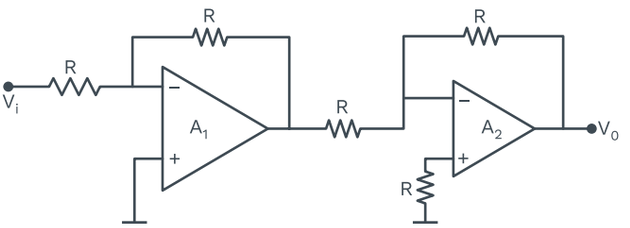


Fig. 9.43 Equivalent Circuit For $V_i > 0$; D_1 is On and D_2 is Off

From the above Fig., we can see that output voltage is equal to input voltage. If negative voltage is applied, then diode D_2 will conduct and diode D_1 will become off. Now equivalent circuit when input voltage is negative is shown in Fig. 9.44.

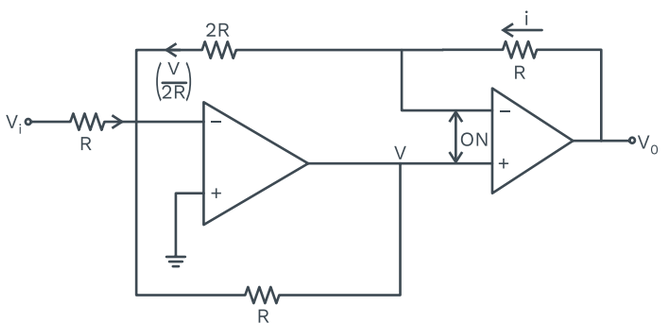


Fig. 9.44 Equivalent Circuit

Let the output voltage of Op-Amp A_1 be 'V'. Since the differential input to A_2 is zero, the inverting input terminals is also at voltage 'V'.

$$\frac{V_i}{R}+\frac{V}{2R}+\frac{V}{R}=0 \qquad (9.118)$$

or, $$V=-\frac{2}{3}V_i \qquad (9.119)$$

The equivalent circuit of (Fig. 9.44) is a non-inverting amplifier as shown in (Fig. 9.45):

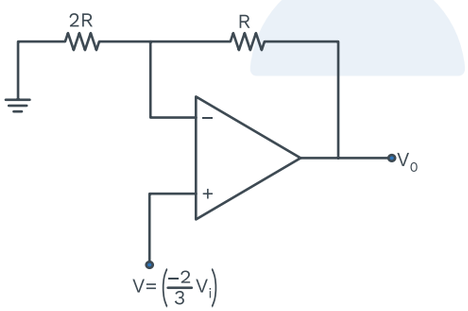


Fig. 9.45 Equivalent Circuit

The output V_0 is:

$$V_0=\left(1+\frac{R}{2R}\right)\left(\frac{-2}{3}V_i\right) \qquad (9.120)$$

$$V_0 = -V_i \qquad (9.121)$$

Hence for $V_i < 0$; the output is positive.

The input and output waveforms are shown in (Fig. 9.46).

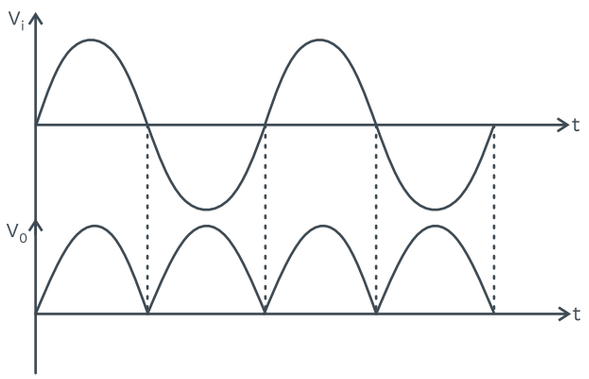


Fig. 9.46 Waveforms of Full Wave Rectifier

The circuit is also called as an absolute value circuit as output is positive even when input is negative.

Summarize the operation of the circuit:

$V_0 = -V_i$ when $V_i < 0V$ (9.122)

$V_0 = V_i$ for $V_i > 0V$ (9.123)

So, $V_0 = |V_i|$ (9.124)

3) Peak value detector:

Definition:

Peak detector is a circuit which is used for determining the peak value of input signal. In peak detector circuit, maximum value of input signal is stored for infinite time duration until reset condition is applied in circuit. The circuits works like:

- Initially peak value is detected at the output.
- Then it stores the value until next peak is detected or reset condition is set.

  Generally peak value of a non-sinusoidal waveform is detected using a peak detector.

Circuit working of peak detector:

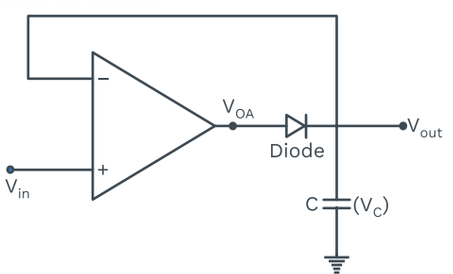


Fig. 9.47 Peak Detector

Fig. 9.47 shows the circuit of basic positive peak detector. It consists of a diode and capacitor along with an Op-Amp as shown above.

Working principle:

- Working principle of the circuit: Peak value of the input waveform is followed and if peak value is detected then it gets stored in terms of voltage in the capacitor.

We will understand through an example. Let us consider a case in which initially V_C voltage is stored in capacitor. When applied voltage is less than V_C ($V_{in} < V_C$) then diode will be in reverse bias and circuit will not work properly, and V_C value is stored in capacitor therefore output voltage will be V_C. Now if input voltage is greater than V_C then diode will become forward biased, and circuit will work as a voltage follower circuit. In this case new value starts accumulating in capacitor and it again reaches to maximum value thus maximum value is stored at capacitor.

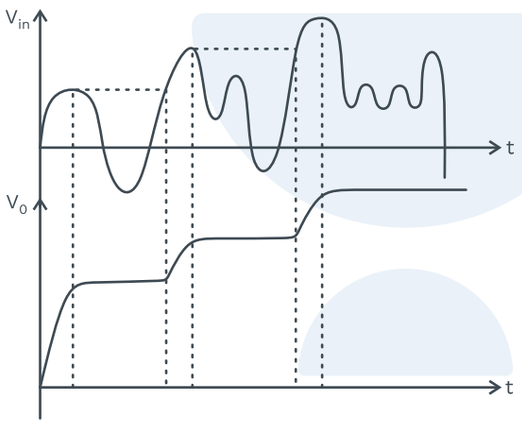


Fig. 9.48 Waveforms of Peak Detector

- We can also construct negative peak detector similar to positive peak detector that will hold the lowest negative value. This can be done by just reversing the polarity of diode in the circuit.

Clipper and clamper circuits:

Clipper circuit: A clipper circuit is an electronic circuit that produces an output by removing a part of input either below or above a reference value. So, we can say output will be same as input except clipped portion and because of this, peak to peak value of output will be less than the peak to peak value of input. One of the main applications of clipper is to remove noise from AC signal.

Classification of clippers:

There are mainly two types of clippers:

- Positive clipper
- Negative clipper

Positive clipper

A Positive clipper circuit is a circuit which clips only positive part(s) of input signal. Fig. 9.49 represents the circuit diagram of a positive clipper.

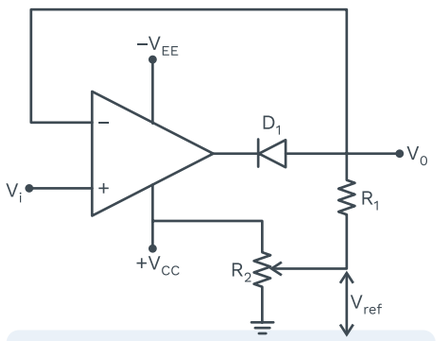


Fig. 9.49 Positive Clipper

As shown in Fig. 9.49, in positive clipper circuit sinusoidal input voltage is applied at non-inverting terminal of Op-Amp. Now value of reference voltage (V_{ref}) can be adjusted by varying the value of resistor R_2. Now we will understand the operation of circuit:

- If the value of applied voltage is less than reference voltage, then diode D_1 will be in forward bias. If diode D_1 is in forward bias, then it will conduct and circuit behaves as a voltage follower circuit, therefore output voltage will be equal to input voltage for ($V_i < V_{ref}$).
- If the value of applied voltage is greater than reference voltage then diode D_1 will be in reverse bias. If diode D_1 is in reverse bias, then it will not conduct, and Op-Amp operates in open loop condition and there is no feedback path present in this condition therefore output voltage will be equal to reference voltage for ($V_i > V_{ref}$).
- Input-output waveform for a negative clipper is shown in Fig. 9.50.

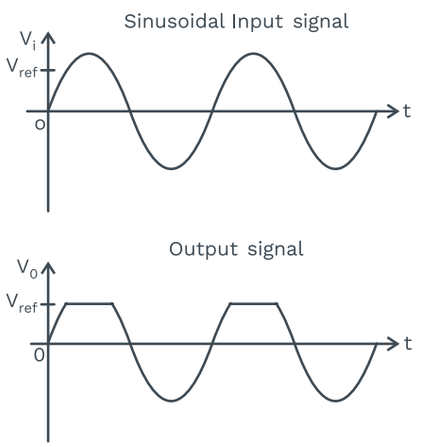


Fig. 9.50 Positive Clipper Waveforms

Negative clipper

A negative clipper circuit is a circuit which clips only negative part(s) of input signal. Negative clipper can be obtained by just reversing the diode in positive clipper circuit and rest all circuit will remain same. Fig. 9.51 represent the circuit diagram of positive clipper.

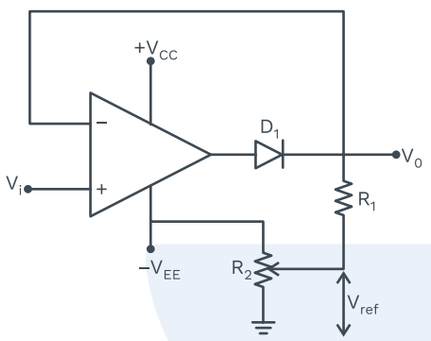


Fig. 9.51 Negative Clipper Circuit

As shown in Fig. 9.51 in negative clipper circuit, sinusoidal input voltage is applied at the non-inverting terminal of Op-Amp. Now value of reference voltage (V_{ref}) can be adjusted by varying the value of resistor R_2. Now we will understand the operation of circuit:

- If the value of applied voltage is greater than reference voltage, then diode D_1 will be in forward bias. If diode (D_1) is in forward bias, then it will conduct and circuit behaves as a voltage follower circuit, therefore output voltage will be equal to input voltage for ($V_i > V_{ref}$).
- If the value of applied voltage is less than reference voltage then diode D_1 will be in reverse bias. If diode D_1 is in reverse bias, then it will not conduct and Op-Amp operates in open loop condition and there is no feedback path present in this condition therefore output voltage will be equal to reference voltage for ($V_i < V_{ref}$).

Input-output waveform for a negative clipper is shown in Fig. 9.52.

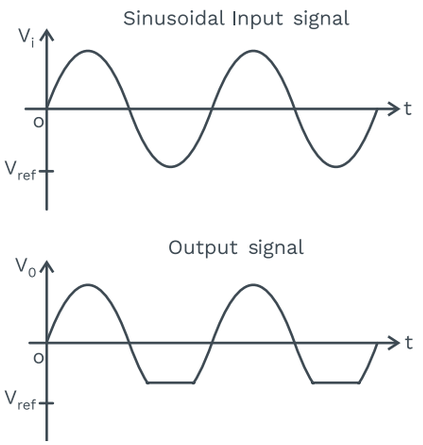


Fig. 9.52 Negative Clipper Waveforms

Clamper circuits: A clamper circuit is an electronic circuit in which output is almost similar to input but with a shift in the DC level. Therefore, we can say that output of clamper circuit is replica of input and due to this peak-to-peak value of output signal is equal to peak-to-peak value of input signal. Clamper circuits are mainly used for either restoring the DC level of input signal at output signal or introducing the DC level to input signal. There are mainly two types of clamper circuits:

- Positive clamper circuits
- Negative clamper circuits

Positive clamper

A positive clamper circuit is a circuit in which shift in output DC level is positive. Therefore, we can say that output of positive clamper is such that the input signal gets shifted vertically by a positive DC value. Circuit diagram for a positive clamper is shown in Fig. 9.53.

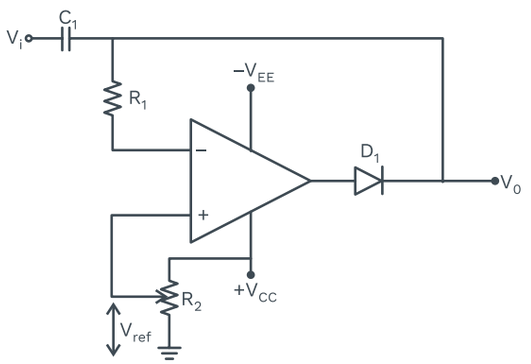


Fig. 9.53 Positive Clamper

- As shown in Fig. 9.53, for positive clamper circuit a sinusoidal input voltage is applied at inverting terminal of Op-Amp by using the network which contains resistor (R_1) and capacitor (C_1). Now DC reference voltage (V_{ref}) is adjusted at non-inverting terminal with the help of resistor R_2. As we can see from Fig. 9.53 that, we have taken value of reference voltage as positive.
- Output of the above circuit will be such that value of output will be equal to sum of input signal voltage (V_i) and reference voltage (V_{ref}). So, we can say output of positive clamper will be such that input sinusoidal voltage will be vertically upward shifted by the value of reference voltage. Input-output waveform for a positive clamper is shown in Fig. 9.54.

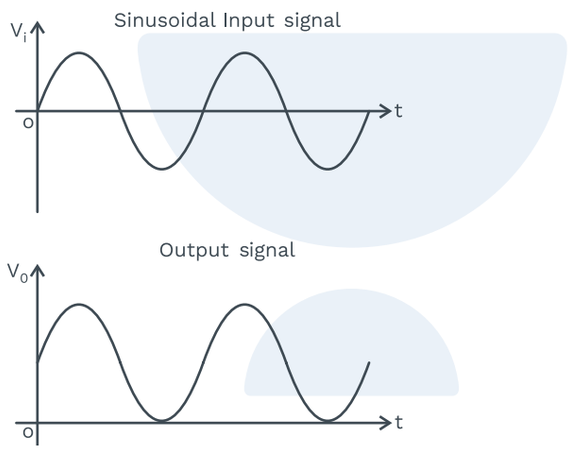


Fig. 9.54 Input-output Waveforms of Positive Clamper

Now we can see from Fig. 9.54 that, output is upward shifted version of input signal and amount of shift will depend upon the value of DC reference voltage applied at non-inverting terminal.

Negative clamper

A negative clamper circuit is a circuit in which shift in output DC level is negative. Therefore, we can say that output of negative clamper is such that the input signal gets shifted vertically by a negative DC value. Circuit diagram for a negative clamper is shown in Fig. 9.55.

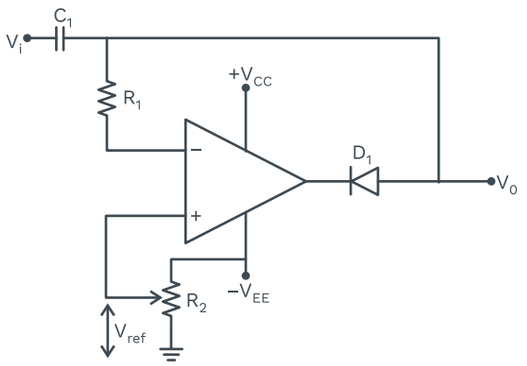


Fig. 9.55 Negative Clamper Circuit

- As shown in Fig. 9.55, for negative clamper circuit a sinusoidal input voltage is applied at inverting terminal of Op-Amp by using the network which contains resistor (R_1) and capacitor (C_1). Now DC reference voltage (V_{ref}) is adjusted at non-inverting terminal with the help of resistor R_2. As we can see from Fig. 9.55 that we have taken value of reference voltage as negative.
- Output of above circuit will be such that value of output will be equal to sum of input signal voltage (V_i) and reference voltage (V_{ref}). Here reference voltage is negative. So, we can say output of negative clamper will be such that input sinusoidal voltage will be vertically downward shifted by the value of reference voltage.

Input-output waveform for a negative clamper is shown in Fig. 9.56.

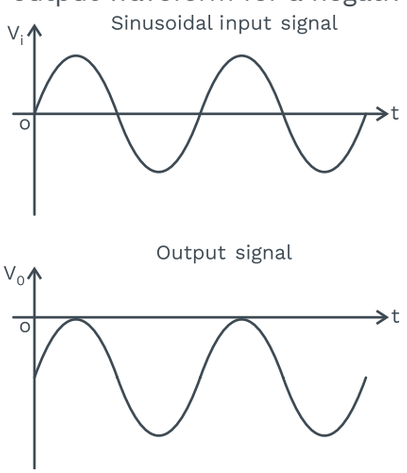


Fig. 9.56 Input-output Waveforms of Negative Clamper

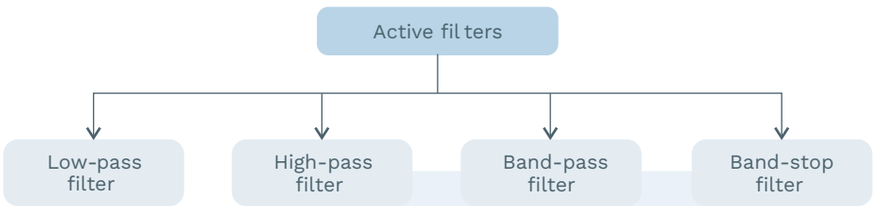
Rack Your Brain

Which Op-Amp doesn't have feedback loop?

Now we can see from Fig. 9.56, that output is the downward shifted version of input signal and amount of shift will depend upon the value of DC reference voltage applied at non-inverting terminal.

Active filters using Op-Amp:

Filter is an electronic circuit which allow to pass certain frequency or reject some part of the frequency. There are two types of filters: Active filter and passive filter. Active filters are those filters which are made by active components like Op-Amp, whereas passive filters are made by passive element like resistor, capacitor and inductor. Active filters also contain passive elements.



Active low-pass filter

Active low-pass filter is an active filter in which low frequency signal is allowed to pass through filter whereas high frequency signals are not allowed to pass (rejects) through filter. Circuit diagram for active low-pass filter is given in Fig. 9.57.

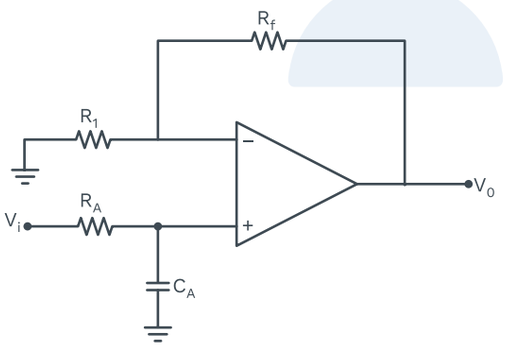


Fig. 9.57 Active Low Pass Filter

- As we can see from Fig. 9.58 a passive low-pass filter is connected to non-inverting terminal of Op-Amp, therefore we can say input to active low-pass filter is a passive low-pass filter.

Analysis of frequency response:

Calculation of transfer function:

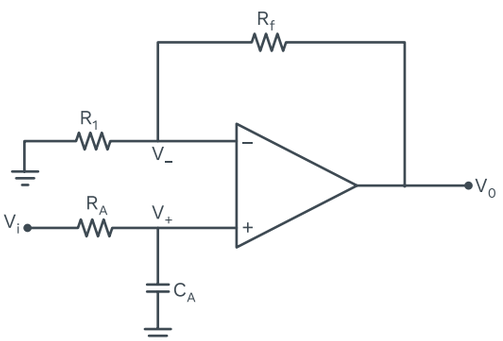


Fig. 9.58 Analysis of Active Low Pass Filter

By voltage division rule,

$$V_+ = \frac{V_i\left(\frac{1}{sC_A}\right)}{R_A + \frac{1}{sC_A}} \quad (9.125)$$

$$V_+ = \frac{V_i}{1 + sC_A R_A} \quad (9.126)$$

Now,

$$V_+ = V_-$$

So, $$V_0 = \left(1 + \frac{R_F}{R_1}\right)V_+ \quad (9.127)$$

$$V_0 = \frac{\left(R_1 + R_F\right)}{R_A} \times \frac{V_i}{1 + sC_A R_A} \quad (9.128)$$

$$\frac{V_0}{V_i} = \left(\frac{K}{1 + sC_A R_A}\right) \quad (9.129)$$

Where,

$$K = \left(1 + \frac{R_F}{R_1}\right) \quad (9.130)$$

From the transfer function,

Cut off frequency for the above filter = $\frac{1}{C_A R_A}$ rad/s

Frequency response of a low-pass filter:

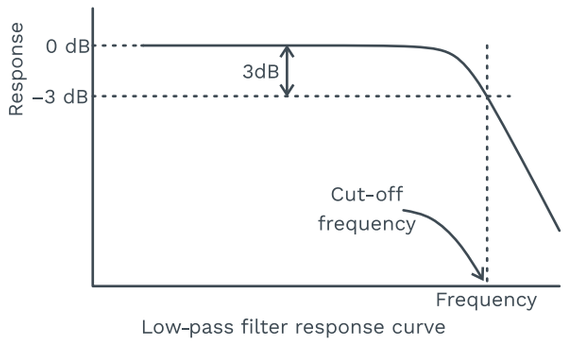


Fig. 9.59 Frequency Response of Low Pass Filter

Values of R_f and R_1 can be adjusted depending upon our requirement of desired gain at output. Now the above circuit will produce a unity gain low-pass filter output if $R_f = 0\ \Omega$ and $R_1 = \infty\ \Omega$.

Active high-pass filter

Active high-pass filter is an active filter in which high frequency signal is allowed to pass through filter whereas low frequency signal are not allowed to pass (rejects) through filter. Circuit diagram for active high-pass filter is given in Fig. 9.60.

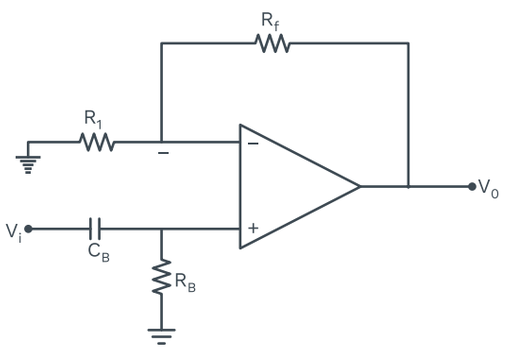


Fig. 9.60 Active High Pass Filter

- As we can see from Fig. 9.60 a passive high-pass filter is connected to the non-inverting terminal of Op-Amp, therefore we can say input to active high-pass filter is a passive high-pass filter.

Frequency response analysis:

Calculation of transfer function:

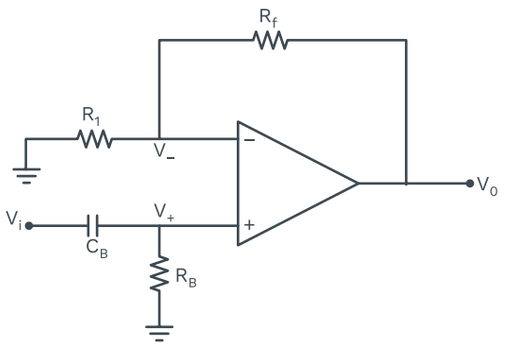


Fig. 9.61 Frequency response Analysis

Now, by voltage division rule:

$$V_+ = \frac{V_i\left(R_B\right)}{\left(\frac{1}{sC_B}\right)+R_B} \tag{9.131}$$

$$V_+ = \frac{V_i\left(sC_B R_B\right)}{1+R_B C_B s} \tag{9.132}$$

Now, as we know, output voltage is given by

$$V_0 = \left(1+\frac{R_F}{R_1}\right)\cdot V_+ \tag{9.133}$$

$$V_0 = \left(1+\frac{R_f}{R_1}\right)\cdot\frac{V_i\left(sC_B R_B\right)}{\left(1+R_B C_B s\right)} \tag{9.134}$$

Now, transfer function,

$$\frac{V_0}{V_i} = \frac{K\left(sC_B R_B\right)}{\left(1+R_B C_B s\right)} \tag{9.135}$$

Where,

$$K = \left(1+\frac{R_f}{R_1}\right)$$

From transfer function,

Cut-off frequency for the above filter $= \frac{1}{R_B C_B}$

Frequency response curve:

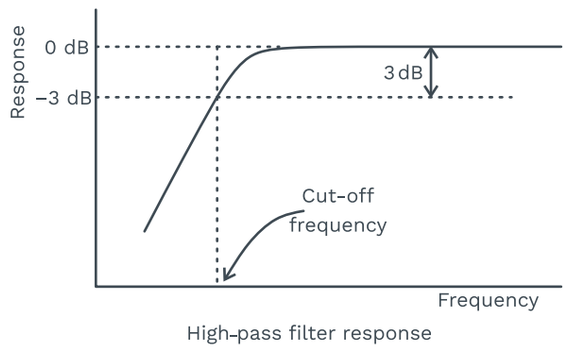


Fig. 9.62 Frequency Response Curve of High Pass Filter

Values of R_f and R_1 can be adjusted depending upon our requirement of desired gain at output. Now the above circuit will produce a unity gain high-pass filter output if R_f = 0 Ω and R_1 = ∞ Ω.

Active band-pass filter

Active band-pass filter is an active filter in which a particular band of frequency signal is allowed to pass through the filter whereas frequencies outside the band are rejected.

Therefore, we can say that an active band-pass filter rejects both high frequency signal as well as low frequency signal. Circuit diagram for active band-pass filter is given in Fig. 9.63.

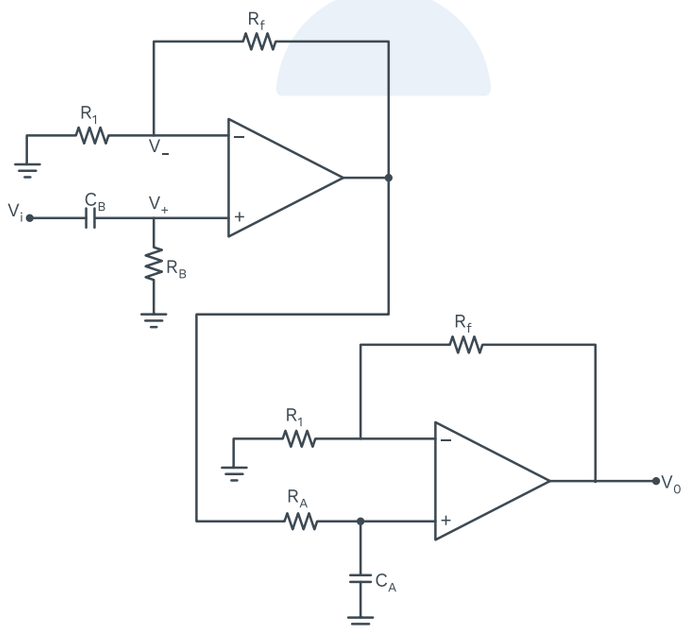


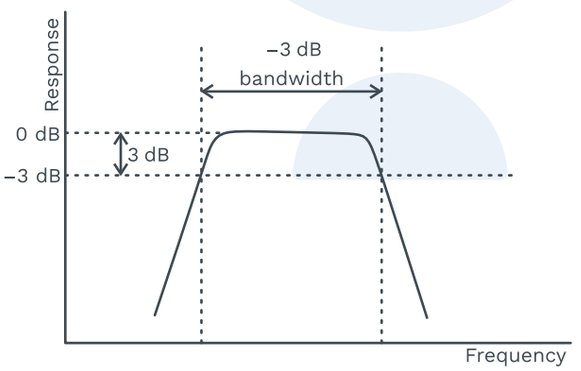
Fig. 9.63 Active Band Pass Filter

As we can see from the diagram of active band-pass filter that there are two parts in circuit diagram. In the first part, there is active high-pass filter whereas in second part active low-pass filter. Output of first stage (active high-pass filter) acts as input to the second stage (active low-pass filter). Therefore we can say that high-pass filter and low-pass filter are cascaded in such a way that it passes a particular range of frequency and rejects other frequency.

In the first stage is active high-pass filter which allow the frequencies greater than lower cut off frequency of active band-pass filter. Therefore, lower cut off frequency is decided by resistor (R_B) and capacitor (C_B). To obtain the precise response, value of resistor and capacitor is chosen properly.

In the second stage, it is the active low-pass filter which allow the frequencies lower than higher cut off frequency of active band pass filter. Therefore, higher cut off frequency is decided by resistor (R_A) and capacitor (C_A). To obtain the precise response, value of resistor and capacitor is chosen properly.

Frequency response curve:



Generic band pass filter response

Fig. 9.64 Frequency Response of Active Band Pass Filter

Active band-stop filter:

Active band-stop filter is an active filter in which particular band of frequency signal is rejected through filter whereas frequency other than the particular band is allowed to pass through filter. Therefore, we can say that an active band-stop filter is that which allow both high frequency signal as well as low frequency signal. Circuit diagram for active band-stop filter is given in Fig. 9.65.

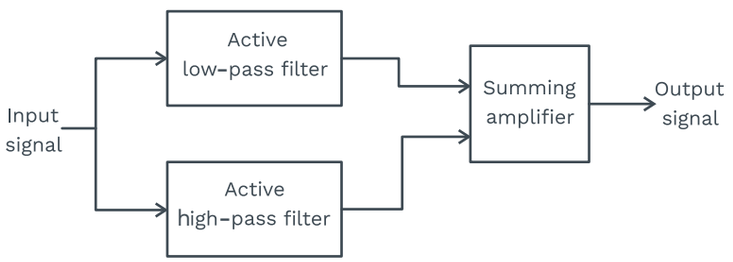


Fig. 9.65 Block Diagram of Active Band Stop Filter

As we can see from the block diagram of active band-stop filter that there are two blocks in first stage in circuit diagram. In one block there is active high-pass filter whereas in another block active low-pass filter. Output of two blocks acts as an input to second stage. Second stage is the summing amplifier and output of second stage is sum of amplified version of active low-pass filter as well as active high-pass filter. In this, the output of summing amplifier will be the output of active band-stop filter if we choose such that the cut-off frequency of low-pass filter is lower than cut-off frequency of high-pass filter. Circuit diagram of active band-stop filter is shown in Fig. 9.65. The circuit diagram of an active band-stop filter is shown in Fig. 9.66.

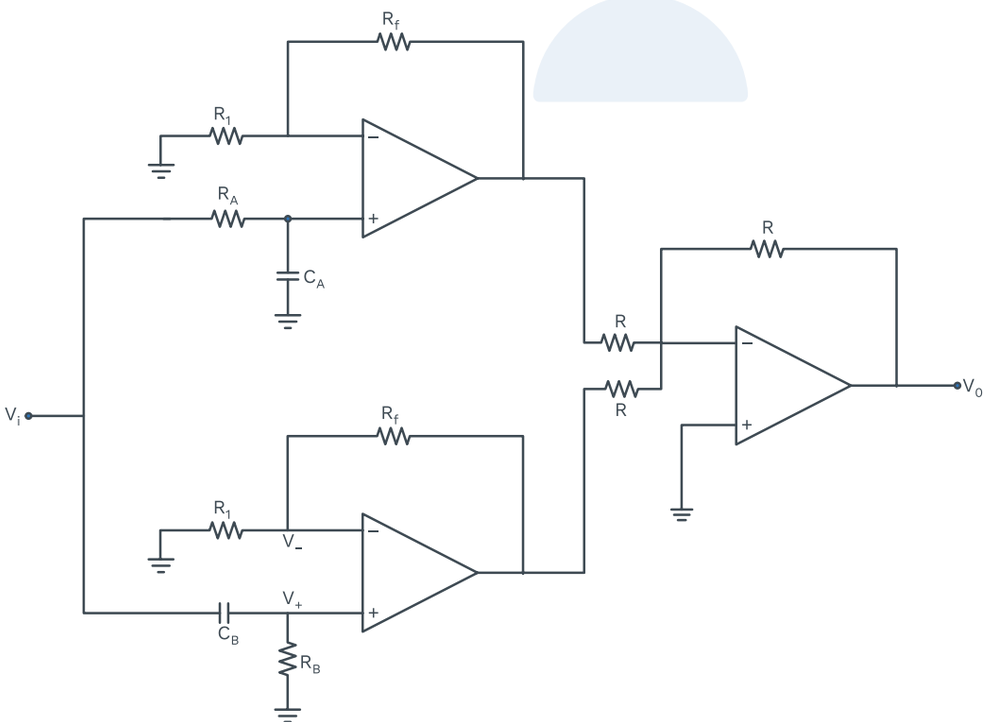


Fig. 9.66 Circuit Diagram of an Active Band Stop Filter

As we can see from Fig. 9.66 and Fig. 9.67 that by replacing equivalent blocks to equivalent circuit, Active band-stop filter can be obtained.

Frequency response curve:

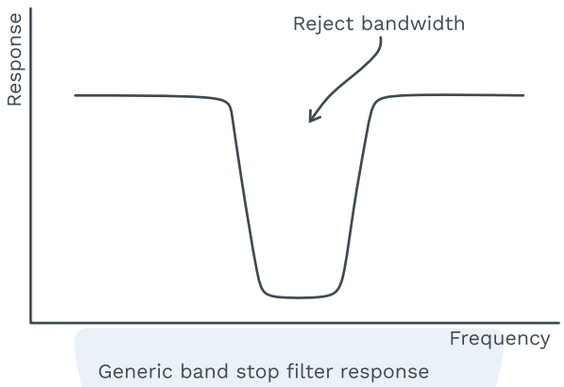
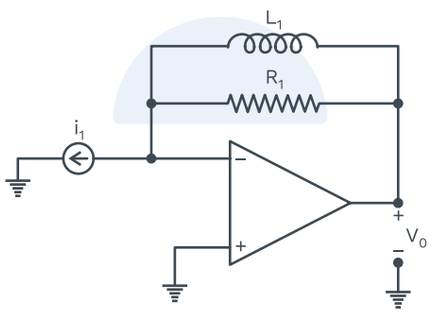


Fig. 9.67 Frequency Response Curve

Example 30:

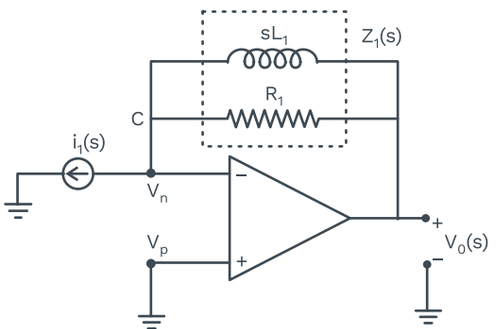
Given below circuit is a filter which is constructed using Op-Amp. The output voltage is V_0 and input current is i_1. What type of filter it is?



a) Band-stop filter **b)** Low-pass filter **c)** Band-pass filter **d)** High-pass filter

Sol:

Given circuit



Here

$V_P = V_n = 0$

KCL at V_n

$$\frac{V_n - V_0(s)}{Z_1(s)} = -i_1(s)$$

$$\Rightarrow -\frac{V_0(s)}{Z_1(s)} = -i_1(s)$$

$$\Rightarrow H(s) = \frac{V_0(s)}{i_1(s)} = z_1(s)$$

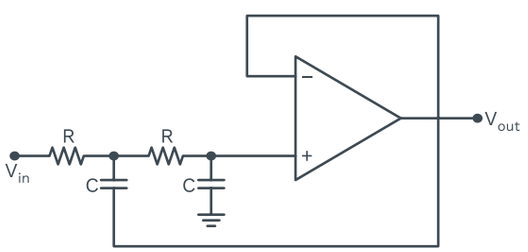
$$\Rightarrow H(s) = \frac{R_1 \cdot sL_1}{R_1 + sL_1}$$

$$\Rightarrow H(s) = \frac{sL_1}{1 + s\left(\frac{L_1}{R_1}\right)}$$

On comparing it with standard equation $H(s) = k\frac{s}{1+sT_1}$, it is apparent that it is a high-pass filter.

Example 31:

Given below circuit is a filter which is constructed using Op-Amp. The output voltage is V_{Out}. What type of filter it is?



a) Band-stop filter
b) Low-pass filter
c) Band-pass filter
d) High-pass filter

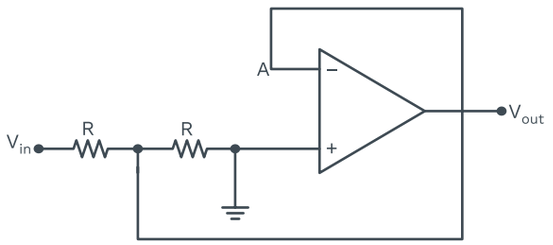
Sol:

The given circuit is a second-order Butterworth filter.

At $\omega = \infty, X_C = \frac{1}{\omega C} = 0$

Both capacitors will be short circuited.

Redraw the circuit,



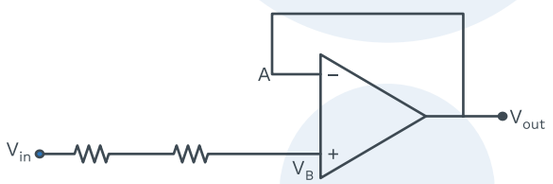
Now, $V_{out} = V_A$

$V_A = V_B = 0$

$V_{out} = 0$

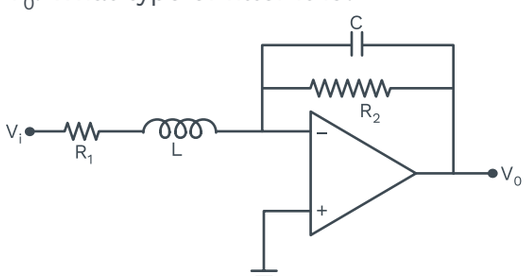
At $\omega = 0, X_C = \infty$

Both capacitors will be open circuit.
Redraw the circuit,



$V_{out} = V_A$
$V_A = V_B$
or, $V_{out} = V_B$ (Non-zero value)
Therefore, given circuit is a low-pass filter.

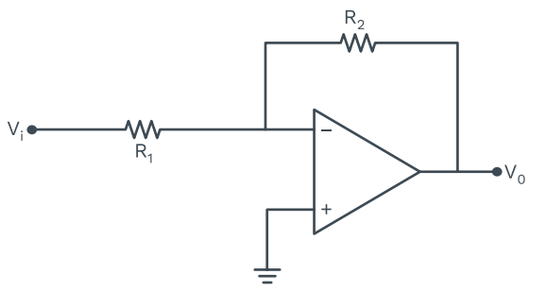
Example 32:
The below given circuit is a filter which is constructed using Op-Amp. The output voltage is V_0. What type of filter it is?



a) Band-stop filter
b) Low-pass filter
c) Band-pass filter
d) High-pass filter

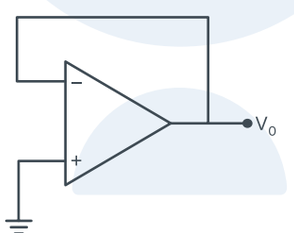
Sol:

At $\omega = 0$, $X_L = 0$ and $X_C = \infty$
Redraw the circuit accordingly



$$V_0 = \frac{-R_2}{R_1} V_i$$

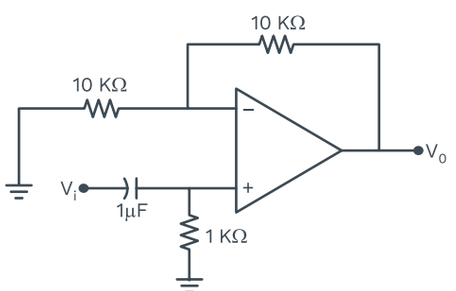
At $\omega = \infty$, $X_L = \infty$ and $X_C = 0$
Redraw the circuit accordingly



$V_0 = 0$
Therefore given circuit will act as a low-pass filter.

Example 33:

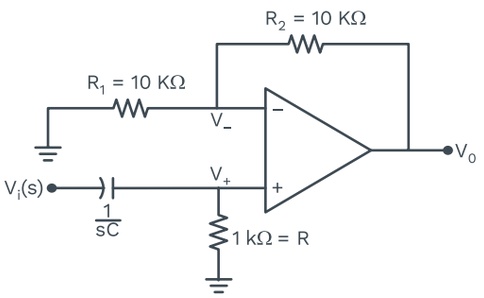
The below given circuit is a filter which is constructed using Op-Amp. The output voltage is V_0. What type of filter it is and cut-off frequency for filter will be?



a) High-pass, 1000 rad/s
b) Low-pass, 1000 rad/s
c) High-pass, 10 rad/s
d) Low-pass, 10 rad/s

Sol:

In s-domain,



From the circuit (in s-domain)

$$V_+(s) = \frac{R}{R + \frac{1}{sC}} \cdot V_i(s)$$

$$V_+(s) = \frac{sRC}{1 + sRC} V_i(s)$$

The circuit is a non-inverting amplifier

$$\therefore V_0(s) = \left(1 + \frac{R_2}{R_1}\right) V_+(s)$$

$$\Rightarrow V_0(s) = (1+1)V_+(s)$$

$$\Rightarrow V_0(s) = 2 \cdot \frac{sRC}{1 + sRC} V_i(s)$$

$$\frac{V_0(s)}{V_i(s)} = H(s) = \frac{s(2RC)}{1 + sRC}$$

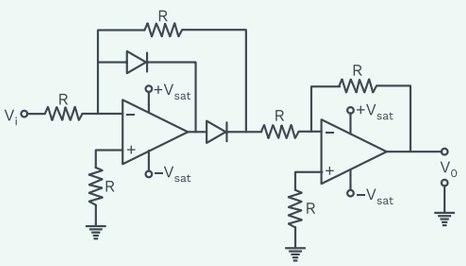
Here, H(s) represents the transfer function of a HPF. The cut-off frequency of a HPF is given as

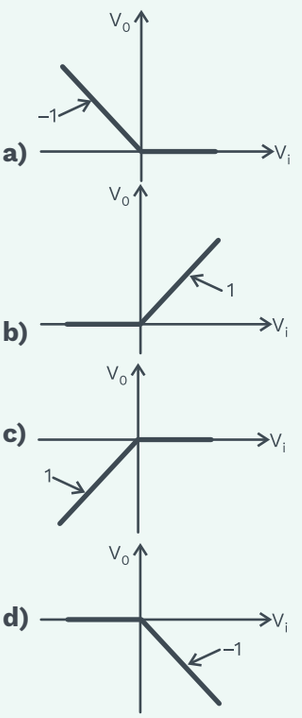
$$\omega_c = \frac{1}{RC} = \frac{1}{1000 \times 1 \times 10^{-6}}$$

$$= 1000 \text{ rad/s}$$

Previous Years' Question

The transfer characteristics of the op-amp circuit shown in the figure. is:





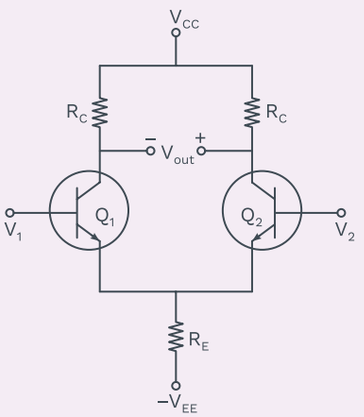
Chapter Summary

Differential amplifier

- Here,

$V_{out} = A(V_1 - V_2)$

Where, V_{out} = Voltage between the collectors



$$A = \frac{R_C}{R_E}$$

where, R_E = Emitter resistance
V_1 = Non-inverting input voltage
V_2 = Inverting input voltage

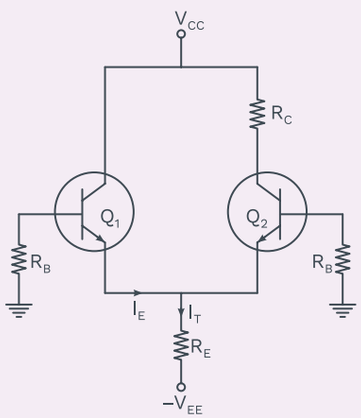
DC analysis of a differential amplifier

- Emitter bias current

$$I_E = \frac{V_{EE} - V_{BE}}{2R_E}$$

- Tail current

$$I_T = \frac{V_{EE} - V_{BE}}{R_E}$$



AC analysis of a differential amplifier

1) Non-inverting input

AC emitter current

$I_E = \frac{V_1}{2R_E}$ where, R_E = emitter resistance

AC output voltage is $V_{out} = I_C R_C = \frac{V_1}{2R_E} R_C$

Voltage gain for non-inverting input is $\frac{V_{out}}{V_1} = \frac{R_C}{2R_E}$

2) Inverting input

AC emitter current is $I_E = \frac{V_2}{2R_E}$

AC output voltage is $V_{out} = -I_C R_C = -\frac{V_2}{2R_E} R_C$

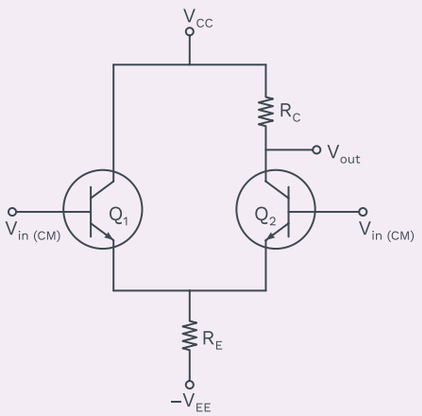
Voltage gain for inverting input is $\frac{V_{out}}{V_2} = \frac{R_C}{2R_E}$

Differential voltage gain $A = \frac{R_C}{2R_E}$

Input impedance $R_{in} = 2\beta R_E$

Common mode voltage gain

$$A_{CM} = \frac{V_{out}}{V_{in(CM)}} = -\frac{R_C}{r_e + 2R_E} \cong -\frac{R_C}{2R_E}$$



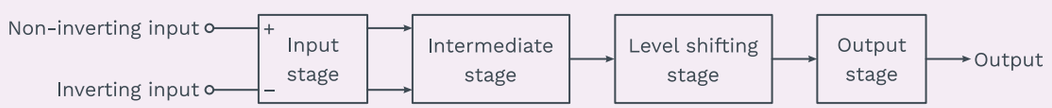
$$CMRR = \frac{A}{-A_{CM}}$$

where, $A_{CM} \rightarrow$ Common mode gain

Note: $R_E >> r_e$

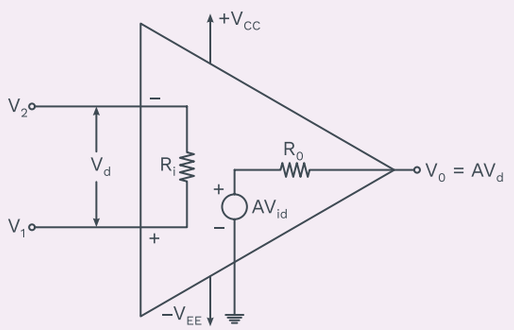
Operational amplifier

An Op-Amp is basically a very high gain, direct-coupled amplifier with high input impedance and low output impedance.



Equivalent circuit of an Op-Amp

Op-Amp amplifies the difference between the two input signals applied at non-inverting and inverting input terminals.



where,

Difference input voltage, $V_d = V_1 - V_2$

A = Open-loop gain of an Op-Amp

Output voltage $V_0 = AV_d$

R_i = Differential input resistance

R_0 = Output resistance

Properties of an op-amp

Parameters	Ideal value	Practical value
Voltage gain	∞	10^6
Input resistance	∞	10^6 Ω or 1 MΩ
Output resistance	0	10 Ω to 100 Ω
Bandwidth	∞	10^6 Hz or 1 MHz
CMRR	∞	10^6 or 120 dB
Slew rate	∞	80 V/μ s

- **Slew rate:** Maximum rate of change of output voltage per unit time under large signal conditions.

$$SR = \left.\frac{dV_0}{dt}\right|_{max} \quad V/\mu - s.$$

For input $V_m \sin \omega t$

$$f_{max} = \frac{SR}{2\pi A_{CL} V_m}$$

where,

SR = Slew rate

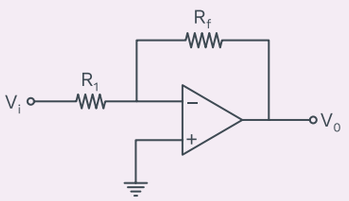
f_{max} = Maximum frequency of operation

A_{CL} = Closed-loop gain of an Op-Amp

Applications of an Op-Amp:

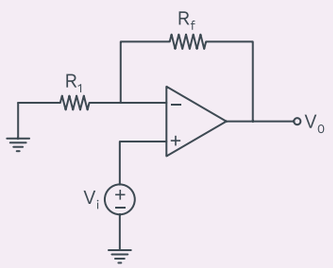
- Linear Op-Amp circuits:

1) Inverting amplifier



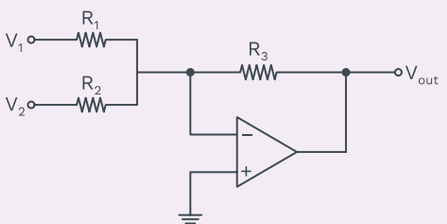
$$\text{Gain, } A_V = -\frac{R_f}{R_1}$$

1) Non-inverting amplifier



$$\text{Gain, } A_V = \left(1+\frac{R_f}{R_1}\right)$$

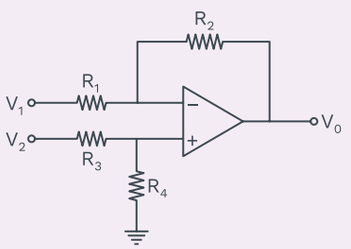
3) Summing amplifier



Output voltage, $V_{out} = \frac{R_3}{R_1}V_1 - \frac{R_3}{R_2}V_2$

Operational amplifier

- Differential amplifier:



$$V_0 = -\frac{R_2}{R_1}V_1 + \left(\frac{R_4}{R_3 + R_4}\right)\left(1 + \frac{R_2}{R_1}\right)V_2$$

If $\frac{R_2}{R_1} = \frac{R_4}{R_3}$

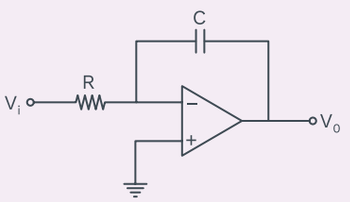
Then, $V_0 = \frac{R_4}{R_3}\left[V_2 - V_1\right] = \frac{R_2}{R_1}\left(V_2 - V_1\right)$

Note: In this case when $\frac{R_2}{R_1} = \frac{R_4}{R_3}$

$A_c = 0 \rightarrow$ Common mode gain and CMRR = ∞

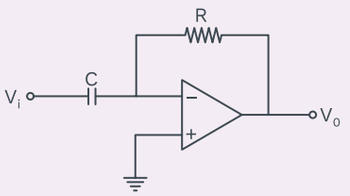
Non-linear Op-Amp circuits:

- **Integrator:**



$$V_0 = -\frac{1}{RC}\int V_i\, dt + V_C\left(0^+\right)$$

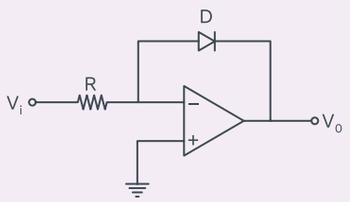
- **Differentiator:**



$$V_0 = -RC\frac{dV_i}{dt}$$

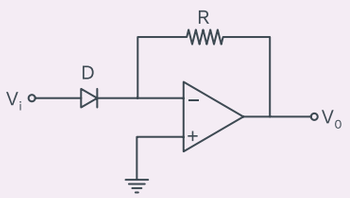
- **Logarithmic amplifier:**

$$V_0 = -\eta V_T \ln\left[\frac{V_i}{I_0 R}\right]$$



where,
η = Recombination factor
V_T = Thermal voltage
I_0 = Reverse saturation current of diode

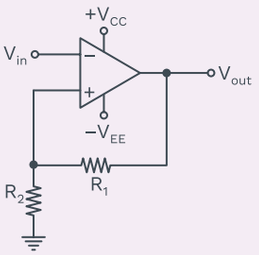
- **Antilog amplifier:**

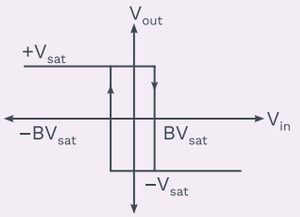


$$V_0 = -I_0.R\,\text{antilog}\left(\frac{V_i}{\eta V_T}\right)$$

where, I_0 = Reverse saturation current of diode
η = Recombination factor
V_T = Thermal voltage

- **Schmitt trigger:**





$$V_{Th} = \frac{R_1}{(R_1 + R_2)} V_H$$

$$V_{TL} = \frac{R_1}{(R_1 + R_2)} V_L \qquad \text{(where } V_L << V_H\text{)}$$

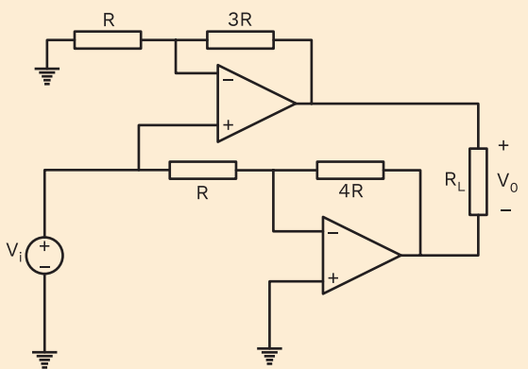
The complete transfer characteristics of a Schmitt trigger shows a hysteresis effect. The width of hysteresis is the difference between the two crossover voltages V_{Th} and V_{TL}.

Op-Amp as a multivibrator

- Square wave generator or an astable multivibrator.
  Time period of output waveform generated is:
  $T = 2\ RC\ \ln(3)$ (for $R_1 = R_2$)
  $$T = 2RC \ln\left[\frac{1+B}{1-B}\right] \quad \left\{\text{where } B = \frac{R_2}{R_1 + R_2}\right\}$$
- Monostable multi-vibrator:
  Width of a pulse generated at output is:
  $$T = RC \ln\left[1 + \frac{R_2}{R_1}\right]$$
  If $R_2 = R_1$,
  then, $T = RC\ \ln(2)$
- Astable multivibrator is a square wave generator.
- Monostable multivibrator is used as pulse stretcher and missing pulse detector.

PRACTICE QUESTIONS

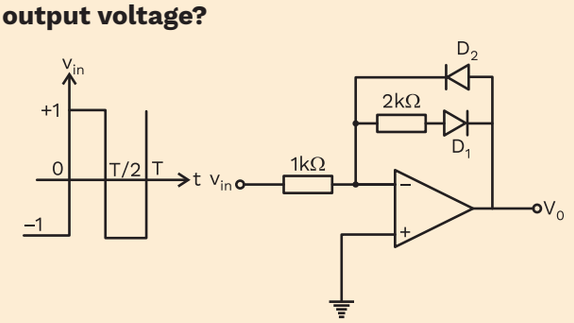
Q1 The gain $\left(\frac{V_0}{V_i}\right)$ of the amplifier circuit shown in the Fig. is:



a) 8 b) 4 c) −4 d) $\frac{3R_L}{R}$

Sol: a)

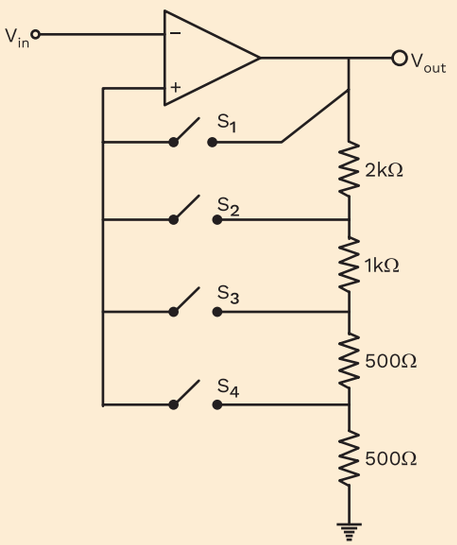
Q2 The below given circuit is an inverting amplifier. Assume diode is ideal and forward resistance of diode is zero whereas forward voltage drop of diode is 0.7 V. Input waveform is shown in given the below figure. What will be the average value of output voltage?



a) −0.7 V b) −1.0 V c) −2.0 V d) −2.7 V

Sol: b)

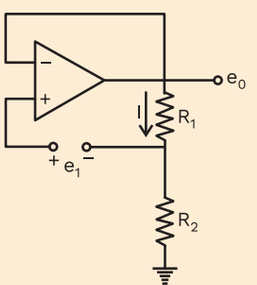
Q3. **What will be the gain of programmable amplifier (as shown below) if switch S_2 is closed?**



a) 0.5 **b) 2** **c) 4** **d) 8**

Sol: **b)**

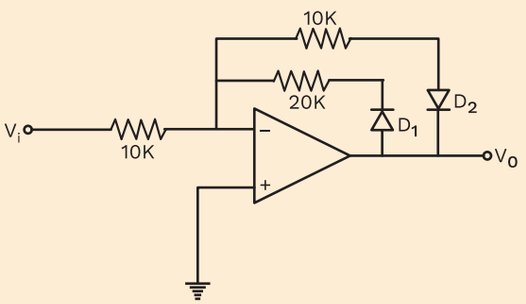
Q4. **Which among the following is the correct relationship for the below given figure?**



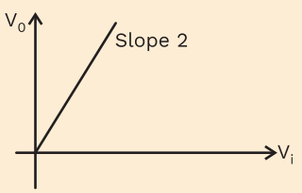
a) $e_0 = e_1$ **b)** $e_0 = \frac{R_2}{R_1} e_1$ **c)** $I = \frac{e_1}{R_2}$ **d)** $I = \frac{e_1}{R_1}$

Sol: **d)**

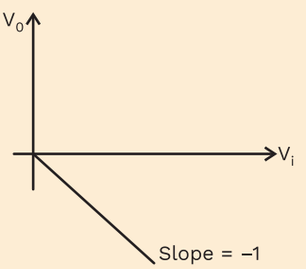
Q5. **The given below circuit id the cut-in voltage of diode and is 0.6 V. Which among the following depicts the correct input-output characteristics of circuit given below?**



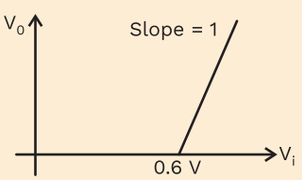
a)



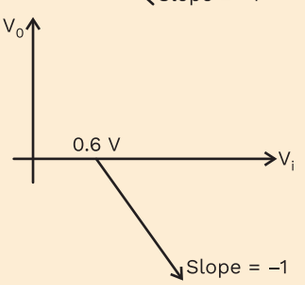
b)



c)

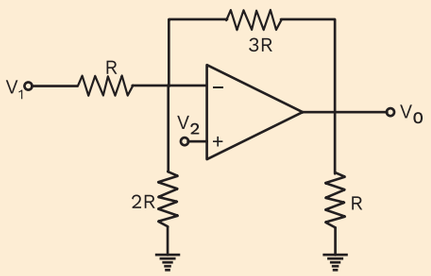


d)



Sol: **d)**

Q6. **Find the value of output voltage if Op-Amp in the below given circuit is ideal?**



a) $\frac{5}{2}V_1 - 3V_2$

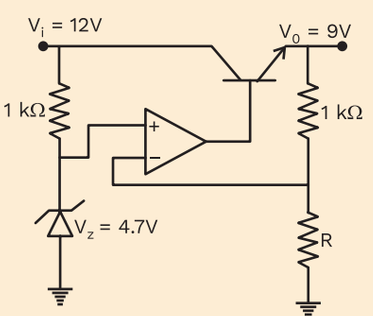
b) $2V_1 - \frac{5}{2}V_2$

c) $-\frac{3}{2}V_1 + \frac{7}{2}V_2$

d) $-3V_1 + \frac{11}{2}V_2$

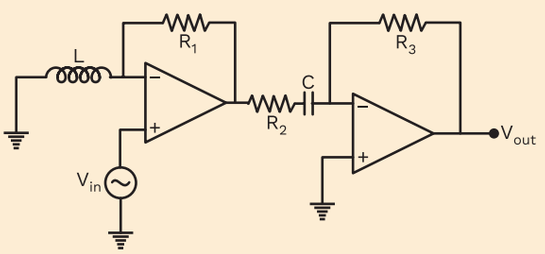
Sol: d)

Q7. **The below given circuit is a voltage regulator circuit. Assume Op-Amp is ideal and BJT has V_{BE} = 0.7 V, β = 100 and zener voltage is 4.7 V. If the output voltage is 9 V then find the value of resistance (R in Ω).**



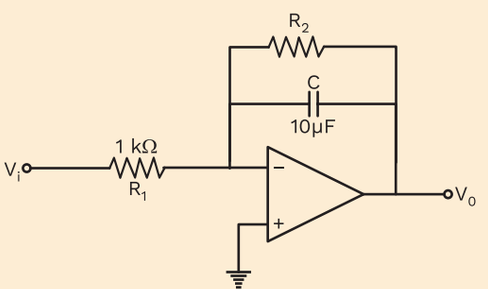
Sol: 1093.02

Q8. **The given below circuit assume Op-Amp is ideal. If input voltage applied is V_{in} = cos (10^6t) then what will be the overall voltage gain (V_{out}/V_{in})? Assume R_1 = R_2 = R_3 = 1 Ω, L = 1 μH and C = 1 μF.**



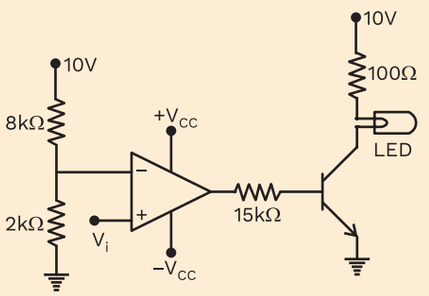
Sol: **–1**

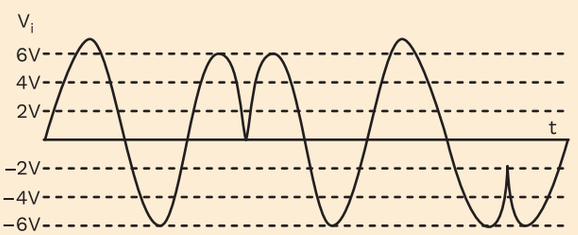
Q9. **In the given below circuit, assume that Op-Amp is ideal and circuit acts as low-pass filter. Find the value of resistance (R_2) if cut-off frequency of low-pass filter is 5 kHz.**



Sol: **3.18**

Q10. **The below given circuit is a voltage regulator circuit. Assume Op-Amp is ideal and BJT has V_{BE} = 0.7 V, β = 100 and V_{LED} = 1.5 V, V_{CC} = 10 V and $-V_{CC}$ = –10 V. If at non-inverting terminal of Op-Amp peak voltage 8 V is applied then how many times LED will glow?**





Sol: 3